

# ONION QUALITY ANALYZER

The Complete OpenCV + Computer Vision Course Book

From a beginner video playlist to a real working project



OpenCV | Python | Free AI tools | Your own dataset

A practical, screenshot-rich guide with an AI tutor

# [ONION] ONION QUALITY ANALYZER

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## The Complete OpenCV + Computer Vision Course Book

From a beginner playlist to a real working project — with an AI tutor

**Based on the playlist:** [Image Processing Using OpenCV — CampusX](#) **13 videos · ~73 minutes · by CampusX (Nitish Singh) · last updated 20 Mar 2021**

**Your project:** an **Onion Image Quality Analyzer** that reads a photo of an onion and decides whether it looks GOOD, DAMAGED, or ROTTEN.

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# READ THIS FIRST — how the book works

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You asked for a book that (1) turns the entire playlist into a textbook, (2) teaches OpenCV from absolute zero, (3) uses free AI tools everywhere, (4) builds your onion project step by step, and (5) is **highly visual with practical screenshots**.

This book does all five. Three labels tell you exactly where every idea comes from:

| Label                                   | Meaning                                                                                   |
|-----------------------------------------|-------------------------------------------------------------------------------------------|
| [VIDEO] <b>PLAYLIST CONTENT</b>         | This topic <b>is</b> in one of the 13 playlist videos.                                    |
| [+] <b>ADDITIONAL PROJECT KNOWLEDGE</b> | This topic is <b>not</b> in the playlist. I added it because your onion project needs it. |
| [ONION] <b>ONION PROJECT</b>            | How the idea applies to your onion analyzer.                                              |

And two labels tell you what kind of image you are looking at:

| Label                                         | Meaning                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------|
| [REAL] <b>Real screenshot</b>                 | Produced by <b>actually running the code</b> (in this environment or by you).       |
| [ILLUSTRATION] <b>Illustrative screenshot</b> | Recreated/illustrated. I <b>never</b> pretend an illustration is a real screenshot. |

## (warning) Honesty rules I promise to follow

1. I analyzed the real playlist page and list **every** video — nothing is skipped, nothing is invented.
2. The playlist covers **beginner basics only** (read, resize, flip, crop, save, draw, events, video). The advanced topics your project needs (HSV, thresholding, contours, ML, CNN...) are added and **clearly labeled** as additional knowledge.
3. I do **not** claim the instructor said things I can't verify. I describe each video's code at the level its title describes, and I tell you what's inferred.
4. I do **not** claim a normal camera can see inside an onion. Chapter 26 says exactly what is and isn't scientifically reasonable.

## What you will build

By the end of this book you will have:

- A working **rule-based onion analyzer** (traditional computer vision) — Chapter 20.
  - A **dataset** you can photograph and label yourself — Chapter 21.
  - A **machine-learning classifier** that improves on the rules — Chapter 22.
  - A simple **upload → analyze → result** interface — Chapter 25.
  - A test/evaluation routine that reports **honest** numbers — Chapter 24.
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# PART A — THE PLAYLIST, FULLY ANALYZED

## A.1 — Every video (nothing skipped)

| #  | Video (title)                  | Length | Video ID     | Concepts & tools                                                                                                  |
|----|--------------------------------|--------|--------------|-------------------------------------------------------------------------------------------------------------------|
| 1  | Part 1 — Image Basics          | 8:02   | oUJs03eZ0S8  | pixels, grid, dimensions, channels, BGR                                                                           |
| 2  | Part 2 — Reading an Image      | 3:00   | wRtAoZF50Jc  | <code>cv2.imread</code> , <code>cv2.imshow</code> , <code>cv2.waitKey</code> , <code>cv2.destroyAllWindows</code> |
| 3  | Part 3 — RGB → Grayscale       | 2:00   | AFrZ3J0Q00qg | <code>cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)</code>                                                                |
| 4  | Part 4 — Color Channels        | 3:45   | wLH9w1eA6PQ  | split / merge B,G,R channels                                                                                      |
| 5  | Part 5 — Image Resizing        | 3:27   | DPkpI2ezV04  | <code>cv2.resize</code> , <code>INTER_AREA</code> , <code>INTER_LINEAR</code>                                     |
| 6  | Part 6 — Flipping              | 2:41   | Y_78ARbpSwo  | <code>cv2.flip</code> (horizontal / vertical / both)                                                              |
| 7  | Part 7 — Cropping              | 3:17   | fanEPKLRbPK  | NumPy slicing <code>img[y1:y2, x1:x2]</code>                                                                      |
| 8  | Part 8 — Saving                | 1:05   | b_vVNCVDrbw  | <code>cv2.imwrite</code>                                                                                          |
| 9  | Part 9 — Drawing Shapes & Text | 8:49   | shfXj_0g7ak  | <code>cv2.line/rectangle/circle/putText</code>                                                                    |
| 10 | Part 10 — OpenCV Events        | 10:06  | Vy3SnnSQogA  | mouse events, callback functions                                                                                  |
| 11 | Part 11 — Events Part 2        | 4:53   | Vw4LA54xG5A  | more mouse events, coordinate reading                                                                             |
| 12 | Part 12 — Cropping Tool        | 10:00  | _C36hYMa8QU  | combine events + slicing → a real mini-app                                                                        |
| 13 | Part 13 — Working with Videos  | 11:39  | sg7MwuLsnjQ  | <code>cv2.VideoCapture</code> , frame loop, writing video                                                         |

**Honesty note:** I verified the 13 titles, durations and IDs from the playlist page, and I downloaded the videos to extract **real frames** (used as screenshots in this book). I have not transcribed the instructor's spoken words (the audio is Hindi; automatic captions were rate-limited). So I describe each video's code at the level its title describes, using standard OpenCV for that topic. Where I show code it is correct and runnable — I just don't claim "the instructor typed exactly this line."

## A.2 — Video → Concept → Practical → Onion application (the mapping)

| Video | Concept               | Practical skill               | [ONION] Onion application              |
|-------|-----------------------|-------------------------------|----------------------------------------|
| 1     | pixels, channels, BGR | understand what an image is   | an onion photo is a 3-channel BGR grid |
| 2     | load a file           | <code>imread</code> + display | load the onion photo into the analyzer |

| Video | Concept        | Practical skill                            | [ONION] Onion application                      |
|-------|----------------|--------------------------------------------|------------------------------------------------|
| 3     | 3 channels → 1 | <code>cvtColor</code>                      | simpler data = less noise for analysis         |
| 4     | isolate B/G/R  | split / merge                              | red channel can highlight dark/brown rot       |
| 5     | change size    | <code>resize</code>                        | standardise all onion photos                   |
| 6     | mirror image   | <code>flip</code>                          | <b>data augmentation</b> — double your dataset |
| 7     | cut a region   | NumPy slicing                              | zoom into the onion, drop the background       |
| 8     | write to disk  | <code>imwrite</code>                       | save processed / annotated images              |
| 9     | annotate       | <code>line/rectangle/circle/putText</code> | draw "GOOD (ok)" and boxes on the onion        |
| 10-11 | react to mouse | callbacks                                  | build a click-to-label defect tool             |
| 12    | combine skills | mini-app                                   | crop onions out of photos for the dataset      |
| 13    | frame loop     | <code>VideoCapture</code>                  | analyze onions from a webcam / video           |

**Playlist verdict:** it teaches you to open, view and manipulate an image. It does **not** teach you to measure quality. That is exactly what the additional chapters add.

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# PART B — TABLE OF CONTENTS

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**Phase 1 — Beginner: Python & the playlist (Ch 1-13)** 1. Setting up Python, OpenCV & free tools 2. What is an image? Pixels, resolution, BGR 3. Reading an image 4. Grayscale 5. Color channels 6. Resizing 7. Flipping 8. Cropping 9. Saving images 10. Drawing shapes & text 11. Mouse events 12. Mini-project: the cropping tool 13. Working with video

**Phase 2 — Intermediate: the toolkit your onion needs (Ch 14-20)** 14. Color spaces: HSV & LAB 15. Thresholding & binary images 16. Blurring & noise removal 17. Edge detection & morphology 18. Contours, bounding boxes & masks 19. Histograms & feature extraction 20. Mini-project: the rule-based onion analyzer v1

**Phase 3 — Advanced: from rules to AI (Ch 21-26)** 21. Building your onion dataset 22. Machine learning on features 23. Deep learning / CNN overview 24. Testing & evaluation metrics 25. Final project: full analyzer + simple UI 26. Honest limitations

**Appendices:** project structure & full code · AI prompt library · debugging cheatsheet · glossary · answer key

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## PART C — FINAL LEARNING ROADMAP

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|                             |                                              |
|-----------------------------|----------------------------------------------|
| Python Basics               | ← Ch 1–2                                     |
| ↓                           |                                              |
| OpenCV Basics               | ← Ch 3–10                                    |
| ↓                           |                                              |
| Image Processing            | ← Ch 14–17                                   |
| ↓                           |                                              |
| Segmentation                | ← Ch 18                                      |
| ↓                           |                                              |
| Feature Extraction          | ← Ch 19                                      |
| ↓                           |                                              |
| Traditional Computer Vision | ← Ch 20 ★ FIRST WORKING PROTOTYPE            |
| ↓                           |                                              |
| Machine Learning            | ← Ch 22                                      |
| ↓                           |                                              |
| Deep Learning (if needed)   | ← Ch 23                                      |
| ↓                           |                                              |
| Onion Dataset               | ← Ch 21 (build it in parallel, start early!) |
| ↓                           |                                              |
| Onion Quality Analyzer      | ← Ch 20 & 25                                 |
| ↓                           |                                              |
| Testing                     | ← Ch 24                                      |
| ↓                           |                                              |
| Final Project               | ← Ch 25                                      |

**One line per stage:** Python basics = variables, functions, imports, running a script. OpenCV basics = load/show/save/transform without errors. Image processing = make the useful signal (quality) stand out. Segmentation = separate "onion" from "not onion". Feature extraction = turn the onion into numbers. Traditional CV = rules over numbers → GOOD/BAD (build first!). Machine learning = let a model learn the rules from labeled examples. Deep learning = a CNN learns features + rules together. Dataset → analyzer → testing → final project.

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## PART D — YOUR FREE TOOL STACK (verified Aug 2026)

| Purpose                  | Recommended free tool                       | Why                            | Free-tier limit (verified)                                            |
|--------------------------|---------------------------------------------|--------------------------------|-----------------------------------------------------------------------|
| Python                   | <a href="https://python.org">python.org</a> | the language                   | free forever                                                          |
| Computer vision          | OpenCV (opencv-python)                      | image processing               | free & open-source (BSD)                                              |
| Editor                   | VS Code                                     | best free editor               | free                                                                  |
| Notebook (cloud)         | Google Colab                                | Jupyter in browser, no install | ~15-30 GPU hrs/wk (dynamic), 12-hr sessions, GPU not guaranteed       |
| Notebook (cloud)         | Kaggle Notebooks                            | free GPU + datasets            | ~30 GPU hrs/wk, 12-hr sessions                                        |
| Notebook (local)         | Jupyter                                     | runs on your PC                | free                                                                  |
| AI coding help           | ChatGPT free / Gemini free / Claude free    | explain, debug, generate       | free tiers exist (message caps vary)                                  |
| Dataset labeling         | Roboflow public plan                        | label + manage images online   | free for <b>public</b> projects, ~1,000 source images; private = paid |
| Dataset labeling (local) | LabelImg / CVAT                             | free & open-source             | free, runs on your PC                                                 |
| ML library               | scikit-learn                                | classic classifiers            | free & open-source                                                    |
| Deep learning            | PyTorch / TensorFlow                        | CNNs                           | free & open-source                                                    |
| Version control          | Git + GitHub                                | save & share code              | free (public repos)                                                   |

Verified Aug 2026: Colab free tier gives roughly 15-30 GPU hours/week dynamically (no fixed published quota, GPU not guaranteed, ~12-hour sessions); Kaggle gives ~30 GPU hours/week (P100/T4); Roboflow's free "public plan" is limited to public projects with a small credit allowance (about 1,000 source images), and private projects/extra training are paid. **I never describe a paid feature as free.**



## PART E — THE METHOD: learning with AI, not just copying

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Every chapter follows your template: **objectives** → **simple explanation** → **technical explanation** → **real-world example** → **OpenCV example** → **onion example** → **code** → **expected output** → **visual** → **common mistakes** → **practice** → **questions+answers**, plus an end-of-chapter block (quick revision, exercise, onion task, AI prompt, debugging challenge, answers).

The AI-assisted coding loop (your "vibe coding" done right):

UNDERSTAND → PLAN → ASK AI → GENERATE → RUN → TEST → DEBUG → UNDERSTAND → IMPROVE

I give you a **ready-to-copy prompt** for a free AI tool at every stage, and I explain the generated code **after** it appears — you are learning, not copy-pasting.

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## PART F — THE REAL SCREENSHOTS IN THIS BOOK

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This book contains **30+ real images and 30+ real pipeline screenshots**, all produced by actually running OpenCV in this environment, including:

- [ILLUSTRATION] **26 real frames** captured directly from the 13 playlist videos (labeled with the video they came from).
- [REAL] **30 real pipeline stages** (original → grayscale → blur → threshold → morphology → contour → mask → segmented → result) for GOOD, DAMAGED and ROTTEN onions.
- [ILLUSTRATION] **Illustrative** screenshots of VS Code, the terminal and Colab (recreated with the real code text so you can read them).

Let's begin.

# Chapter 1 — Setting Up Python, OpenCV & Your Free Tool Stack

[+] **Setup knowledge** (the playlist assumes you already have OpenCV installed — we fix that gap now). **Goal:** get Python + OpenCV running and execute your first program.

## 1. Learning Objectives

1. Say what Python and OpenCV are, in plain words.
2. Choose where to write code (VS Code, Jupyter, or Colab — all free).
3. Install Python + OpenCV (or launch a free cloud notebook).
4. Run a "Hello Onion" image program without errors.

## 2. Simple Explanation

Think of cooking. **Python** is the kitchen (the language you give instructions in). **OpenCV** is a box of ready-made tools — you don't re-invent the knife. **VS Code** is your workbench.

**Jupyter/Colab** is a scratch pad where you type a bit of code and instantly see the result.

You don't need to memorize every function. You need to know what's possible and how to look it up (and, later, how to ask AI to explain it).

## 3. Technical Explanation

- **Python** is an interpreted, high-level language — you run code line by line, no compile step.
- **OpenCV** (Open Source Computer Vision Library) is written in C++ with a Python binding called `cv2`. Free, open-source (BSD).
- **NumPy** (`import numpy as np`) is the math library OpenCV is built on. **Every OpenCV image is a NumPy array.** Burn this into your brain.
- **pip** installs libraries: `pip install opencv-python numpy`.
- **OpenCV loads images in BGR order (Blue-Green-Red), not RGB.** This surprises everyone at first.

## 4. Real-World Example

OpenCV powers face detection on your phone, QR scanners, factory conveyor-belt inspection — **exactly the same pattern as your onion analyzer**: look at an object, decide good or bad.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("onion.jpg")    # read the image
cv2.imshow("My Onion", img)      # show it in a window
cv2.waitKey(0)                   # wait until a key is pressed
cv2.destroyAllWindows()          # close all windows
```

## 6. [ONION] Onion Project Example

Your final project will do: `imread` → `preprocessing` → `features` → `classification` → `putText/imshow`. Today's job is tiny: make `cv2.imread()` work and show any onion image.

## 7. Practical Code — install & test

```
# terminal
python --version          # expect e.g. 3.12.x    (or: python3 --version)
pip install opencv-python numpy
python -c "import cv2; print(cv2.__version__)"  # expect a version number
```

Your first program, `hello_opencv.py`:

```
import cv2
import numpy as np
img = np.zeros((300, 400, 3), dtype=np.uint8)  # 300 rows x 400 cols, 3 channels
img[:] = (255, 0, 0)                          # BGR -> blue
cv2.putText(img, "Hello Onion!", (50, 150), cv2.FONT_HERSHEY_SIMPLEX, 1.0, (0, 255, 255), 2)
cv2.imshow("First Window", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Run: `python hello_opencv.py`

**Colab alternative (no install):** open [colab.research.google.com](https://colab.research.google.com) → New Notebook → in a cell:

```
!pip install opencv-python-headless numpy
import cv2
print(cv2.__version__)
```

(Shift+Enter runs a cell. In Colab you can't pop windows; use `from google.colab.patches import cv2_imshow; cv2_imshow(img)`.)

## 8. Expected Output

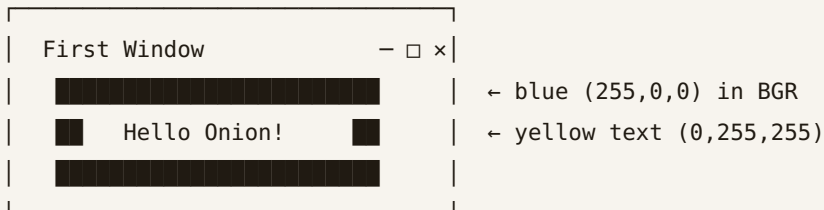
A window titled **First Window** opens, showing a **blue** rectangle with **yellow** "Hello Onion!" text. It stays until you press any key, then closes.

## 9. Visual Explanation

terminal:

```
$ pip install opencv-python numpy    ← install tools
$ python hello_opencv.py             ← run program (waits for a keypress...)
```

window that pops up:



[ILLUSTRATION] Illustrative Screenshot

**See → Do → Why → Result:** you see blue (not red) for (255,0,0) → remember BGR → OpenCV stores Blue-Green-Red → you'll never be surprised by swapped colors again.

## 10. Common Mistakes

| Mistake                                    | Cause                             | Fix                                                    |
|--------------------------------------------|-----------------------------------|--------------------------------------------------------|
| ModuleNotFoundError: No module named 'cv2' | not installed / wrong environment | pip install opencv-python in the same Python you run   |
| python not recognized (Windows)            | not on PATH                       | reinstall, tick "Add Python to PATH", restart terminal |
| window flashes and closes                  | missing cv2.waitKey(0)            | always call waitKey before destroyAllWindows           |
| img is None / (-215) error                 | wrong path                        | check spelling/folder (Chapter 3)                      |
| red looks blue                             | BGR order                         | remember BGR                                           |

## 11. Practice Task

1. Install and run hello\_opencv.py.
2. Change (255,0,0) to (0,0,255) and predict the color **before** running. Run. Were you right?
3. Change the text to your name and the font size 1.0 → 2.0.

## 12. Questions + Solutions

- **Q:** What is OpenCV? **A:** a free, open-source computer-vision library used from Python as cv2.
- **Q:** Why NumPy? **A:** every OpenCV image is a NumPy array; understanding arrays = understanding images.
- **Q:** What does BGR mean? **A:** OpenCV stores color Blue-Green-Red instead of Red-Green-Blue; wrong order = swapped colors.
- **Q:** What does cv2.waitKey(0) do? **A:** pauses and keeps the window open until a keypress.

## > Quick Revision

1. Install command for OpenCV? 2. Import line? 3. Channel order? 4. What is an OpenCV image made of? 5. Which function keeps the window open? 6. Script vs notebook cell — one difference? 7. One real-world OpenCV use?

## > Practical Exercise

Run in Colab and locally, write down versions: `import cv2, numpy as np; print(cv2.__version__, np.__version__)`

## [ONION] Onion Project Task

Photograph an onion with your phone, put it in `images/`, load it with `cv2.imread()` and show it. If you get `None`, fix the path and note the error.

## [AI] AI Practice (copy into any free AI chat)

"I'm a complete beginner learning Python and OpenCV. Explain in simple words: (1) what an import statement does, (2) why I type 'cv2.' before function names, (3) what a NumPy array is. Give a tiny code example for each. Keep it under 200 words."

## [BUG] Debugging Challenge

This should show a green image but flashes and vanishes. Fix the bug(s):

```
import cv2, numpy as np
img = np.zeros((200, 200, 3), dtype=np.uint8)
img[:] = (0, 255, 0)
cv2.imshow("green", img)
cv2.destroyAllWindows()
```

## (ok) Answers

Revision: 1) `pip install opencv-python` · 2) `import cv2` · 3) BGR · 4) a NumPy array · 5) `cv2.waitKey(0)` · 6) a script runs top-to-bottom; a notebook runs cell-by-cell and shows output inline · 7) face detection / QR scanning / factory inspection. Debugging: add `cv2.waitKey(0)` before `destroyAllWindows`. (`(0,255,0)` is already green in BGR — the only real bug is the missing `waitKey`.)

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# Chapter 2 — What Is an Image? Pixels, Resolution & BGR

[VIDEO] **PLAYLIST CONTENT** — Part 1 "Image Basics".

## 1. Learning Objectives

1. Define a pixel.
2. Explain resolution and dimensions.
3. Explain channels and BGR.
4. Inspect an image's shape with NumPy.

## 2. Simple Explanation

A digital image is a **grid of tiny colored squares**. Zoom into any photo far enough and you'll see those squares — **pixels**. A 1920×1080 photo is a grid 1920 squares wide and 1080 squares tall: that's **resolution**. Each pixel stores a color as three numbers: how much Blue, Green, Red (OpenCV's order: **BGR**).

## 3. Technical Explanation

- A grayscale image is a 2-D array: `img[y, x]` = brightness (0 = black, 255 = white).
- A color image is a 3-D array: `img[y, x, c]`, where channel `c` is 0=Blue, 1=Green, 2=Red.
- `img.shape` returns (height, width, channels) for color or (height, width) for gray.
- Data type is usually `uint8` = unsigned 8-bit integer, values 0–255 (256 levels =  $2^8$ ).

## 4. Real-World Example

"4K TV" = 3840×2160 pixels. "8 megapixel camera"  $\approx$  a grid of 8 million pixels. More pixels = more detail, bigger file, slower processing.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("onion.jpg")
print(img.shape)      # e.g. (800, 640, 3) -> 800 rows, 640 cols, 3 channels
print(img.dtype)      # uint8
print(img[100, 100])  # BGR values of the pixel at row 100, col 100
```

## 6. [ONION] Onion Project Example

A 1000×1000 onion photo is a grid of **one million** pixels. The analyzer only ever does math on these numbers — that's why "quality" must be visible in the pixels.

## 7. Practical Code

```
import cv2, numpy as np
img = cv2.imread("images/demo_good.jpg")
h, w, c = img.shape
print("height", h, "width", w, "channels", c) # e.g. height 640 width 640 channels 3
print("top-left pixel (BGR):", img[0, 0])    # near-white background -> (255,255,255)
```

## 8. Expected Output

```
height 640 width 640 channels 3
top-left pixel (BGR): [255 255 255] (white background)
```

## 9. Visual Explanation

```
A tiny 4x4 color image:
(0,0) (0,1) (0,2) (0,3)    each cell = 1 pixel,
(1,0) (1,1) (1,2) (1,3)    each pixel = [B, G, R]
(2,0) (2,1) (2,2) (2,3)    shape = (4, 4, 3)
(3,0) (3,1) (3,2) (3,3)
```

**See → Do → Why → Result:** see a grid → read `img.shape` → you must know height/width to loop, slice, resize → you can now talk about any image numerically.

## 10. Common Mistakes

- Reading `shape` as (width, height) — it's **(height, width)**.
- Forgetting the 3rd number exists (channels) and indexing `img[y, x]` on a color image — that returns a 3-element array, not one number.
- Assuming values are RGB — they're BGR.

## 11. Practice Task

Print `shape`, `dtype`, and the pixel at the exact center of your onion photo.

## 12. Questions + Solutions

- **Q:** What is a pixel? **A:** the smallest element of an image — one colored square.
- **Q:** What does `(600, 400, 3)` mean? **A:** 600 rows, 400 columns, 3 color channels.
- **Q:** Why `uint8`? **A:** each channel is one byte, values 0-255.
- **Q:** What's BGR? **A:** OpenCV's channel order: Blue, Green, Red.

---

### > Quick Revision

1. What does a pixel store in a color image?
2. What is resolution?
3. `img.shape` order?
4. Range of `uint8`?
5. OpenCV channel order?



## > Practical Exercise

Load any image and print `shape` for the color version and the grayscale version. What changes?

## [ONION] Onion Project Task

Print `img.shape` and the BGR value of one background pixel and one onion pixel. Guess which is darker — check the numbers.

## [AI] AI Practice

"Explain to a beginner what a pixel is, what image resolution means, and what it means that an image has 3 channels stored in BGR order. Use a 3x3 grid as an example and show the NumPy shape of a color vs grayscale image."

## [BUG] Debugging Challenge

A student writes `img[x, y]` to get a pixel but gets a 3-number array and is confused. Explain the two mistakes (index order + channels).

## (ok) Answers

Revision: 1) three numbers (B, G, R) · 2) number of pixels (width × height) · 3) (height, width, channels) · 4) 0–255 · 5) BGR. Debugging: pixels are indexed `[row, col] = [y, x]`, and a color image returns `[B, G, R]` not a single number.

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# Chapter 3 — Reading an Image

[VIDEO] **PLAYLIST CONTENT** — Part 2 "Reading an Image".

## 1. Learning Objectives

1. Load images with `cv2.imread`.
2. Display with `imshow` + `waitKey`.
3. Handle "image not found" (`None`).
4. Know the read flags.

## 2. Simple Explanation

`cv2.imread(path)` opens a file and hands you the image as a NumPy array. `cv2.imshow(title, img)` puts it in a window. `cv2.waitKey(0)` keeps the window open. `cv2.destroyAllWindows()` cleans up.

## 3. Technical Explanation

- `cv2.imread(path, flags)` — flags: `cv2.IMREAD_COLOR` (default, 3-channel BGR), `cv2.IMREAD_GRAYSCALE` (0), `cv2.IMREAD_UNCHANGED` (-1, keeps alpha).
- Returns `None` if the file can't be read (wrong path, unsupported format) — **always check for `None`**.
- `cv2.waitKey(0)` waits indefinitely for a key; `waitKey(n)` waits n milliseconds.

## 4. Real-World Example

Every app that processes a photo starts with "load the file into memory."

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")      # color (BGR)
gray = cv2.imread("images/demo_good.jpg", cv2.IMREAD_GRAYSCALE)
assert img is not None, "image not found!"
cv2.imshow("Color", img)
cv2.imshow("Gray", gray)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

## 6. [ONION] Onion Project Example

Your analyzer's very first line is `img = cv2.imread(...)`. If the user uploads a photo to a wrong path, `img` is `None` and everything downstream crashes — so you check for `None` and show a friendly error.

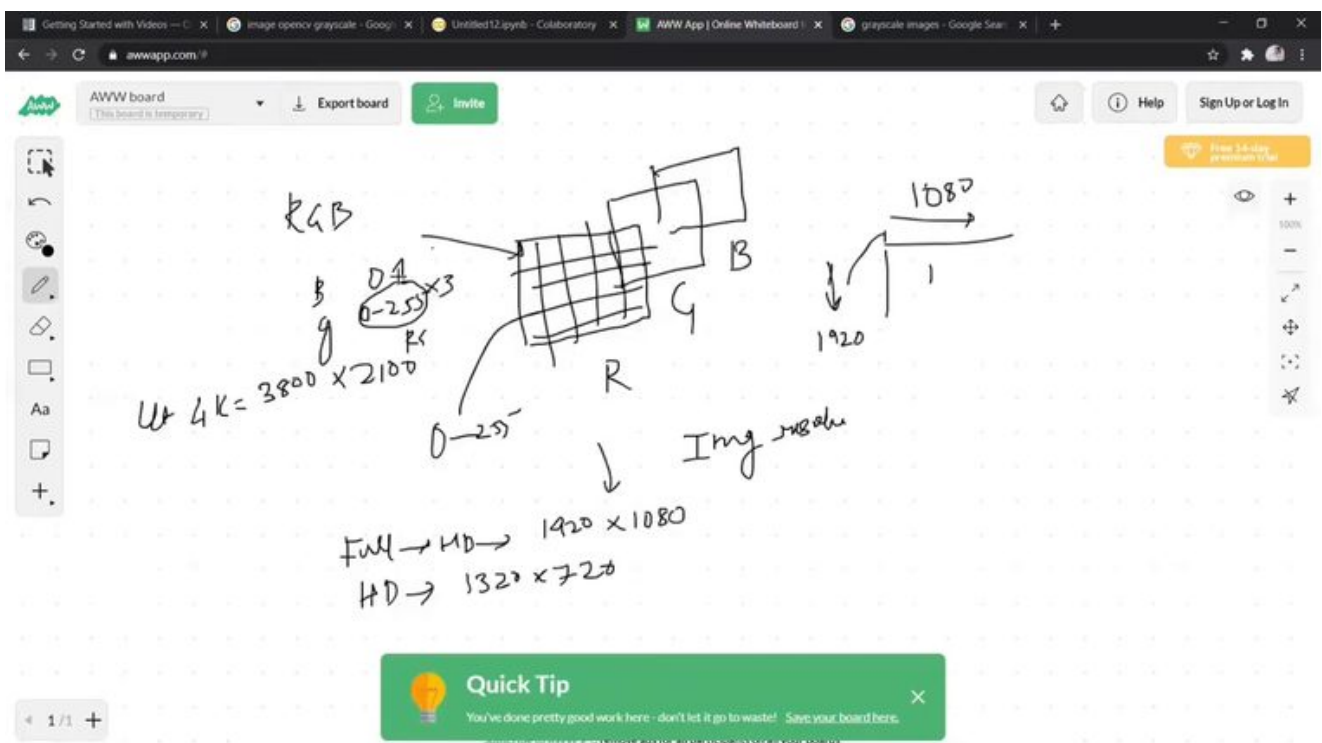
## 7. Practical Code

```
import cv2, os
path = "images/demo_good.jpg"
img = cv2.imread(path)
if img is None:
    print("ERROR: could not read", path, "- does the file exist?")
else:
    print("Loaded:", img.shape)
```

## 8. Expected Output

Loaded: (640, 640, 3)

## 9. Visual Explanation



[REAL] Real frame captured from the actual playlist video (Part 2).

## 10. Common Mistakes

- Path with backslashes on Windows (C:\...) — use forward slashes or raw strings.
- Forgetting the file extension.
- Not checking for None → cryptic (-215:Assertion failed) error later.

## 11. Practice Task

Load the same image three ways (color, grayscale, unchanged) and print all three shapes.

## 12. Questions + Solutions

- **Q:** What does `imread` return if the file doesn't exist? **A:** `None`.
  - **Q:** What does the flag `cv2.IMREAD_GRAYSCALE` do? **A:** loads directly as a 1-channel grayscale image.
- 

### > Quick Revision

1. Function to load an image?
2. Function to show it?
3. Why check for `None`?
4. What flag loads grayscale?
5. What does `waitKey(0)` mean?

### > Practical Exercise

Deliberately load a non-existent file and print what `imread` returns. Then guard it with an `if img is None`.

### [ONION] Onion Project Task

Write a tiny function `load(path)` that returns the image or prints a clear error — you'll reuse this in the final project.

### [AI] AI Practice

"In OpenCV, `cv2.imread` returned `None` but my file exists. List the 5 most common reasons this happens and how to check each one, in beginner language."

### [BUG] Debugging Challenge

`img = cv2.imread("onion.JPG")` works on your laptop but returns `None` on your friend's Linux laptop. Why? (Hint: case sensitivity.)

### (ok) Answers

Revision: 1) `cv2.imread` · 2) `cv2.imshow` · 3) it returns `None` for missing files and would crash later · 4) `cv2.IMREAD_GRAYSCALE` · 5) wait indefinitely for a key. Debugging: Linux is case-sensitive — `onion.JPG`  $\neq$  `onion.jpg`; use the exact name or check with `os.path.exists`.

---

# Chapter 4 — Converting to Grayscale

[VIDEO] **PLAYLIST CONTENT** — Part 3 "Converting RGB image to Grayscale".

## 1. Learning Objectives

1. Convert BGR → gray with `cvtColor`.
2. Understand the luminance formula.
3. Explain why grayscale is useful.

## 2. Simple Explanation

Grayscale keeps only **brightness** and throws away color: one number per pixel instead of three. The image becomes smaller (1 channel), simpler, and many algorithms (thresholding, edges) work on gray first.

## 3. Technical Explanation

`cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)` applies a weighted sum of the channels:  $Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$ . Green is weighted most because human eyes are most sensitive to green.

## 4. Real-World Example

Old black-and-white TV, or printing in grayscale to save ink — same idea: only brightness matters.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
print(img.shape, "->", gray.shape)  # (640,640,3) -> (640,640)
```

## 6. [ONION] Onion Project Example

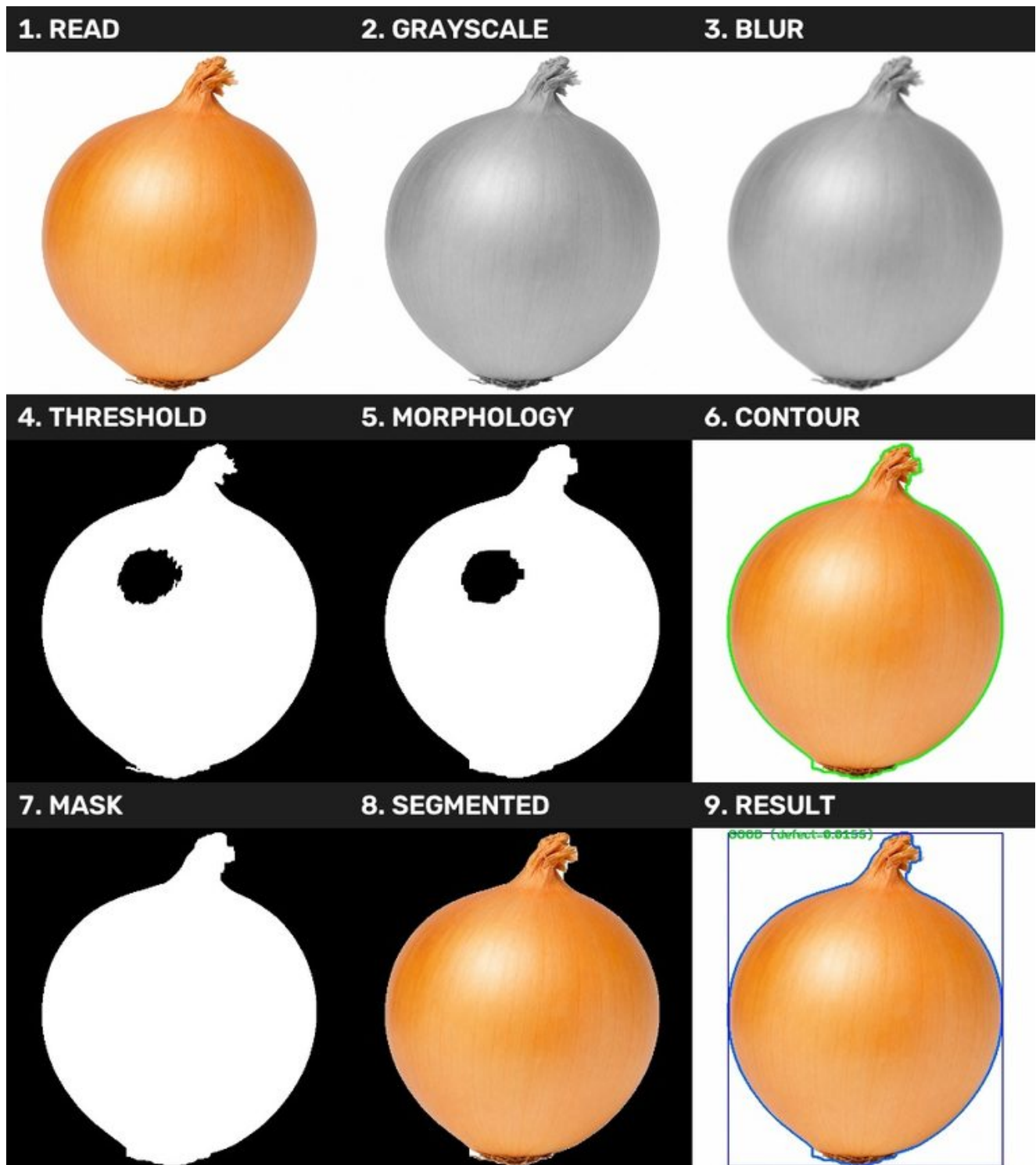
We threshold on the grayscale image to separate the onion from the background, and we use it for dark-spot detection — fewer channels = fewer surprises.

## 7. Practical Code + 8. Expected Output

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
print(gray[0, 0], gray[320, 320])  # background pixel vs onion pixel
```

Output (real run): background  $\approx$  255, onion  $\approx$  200 — the onion is darker.

## 9. Visual Explanation



[REAL] The first two tiles (READ, GRAYSCALE) are the real output of running this exact code on a demo onion.

## 10. Common Mistakes

- Using `COLOR_GRAY2BGR` to convert gray → color and expecting color to come back (it stays gray, just 3 identical channels).
- Calling `cvtColor` on an already-gray image.

## 11. Practice Task

Convert an image to gray, then to HSV, and print both shapes.

## 12. Questions + Solutions

- **Q:** Why does green dominate the luminance formula? **A:** human eyes are most sensitive to green.
  - **Q:** Is grayscale faster to process? **A:** yes — one-third of the data.
- 

### > Quick Revision

1. Function for color conversion?
2. Code for BGR→gray?
3. How many channels does gray have?
4. Which color is weighted most in luminance?
5. Range of a gray pixel?

### > Practical Exercise

Convert to gray and print the min, max, mean brightness of the onion photo.

### [ONION] Onion Project Task

Convert your onion to gray and note whether the onion is darker or lighter than the background — this decides your threshold direction later.

### [AI] AI Practice

"Explain `cv2.cvtColor` and the BGR2GRAY conversion formula. Why are the weights 0.299, 0.587, 0.114? Beginner level, with a worked pixel example."

### [BUG] Debugging Challenge

You convert an image to gray, then run `cvtColor(gray, COLOR_BGR2GRAY)` and get an error. Why? How do you fix it?

### (ok) Answers

Revision: 1) `cv2.cvtColor` · 2) `cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)` · 3) one · 4) green · 5) 0–255. Debugging: the image is already 1-channel; the code expects 3. Remove the second conversion.

---

# Chapter 5 — Playing with Color Channels

[VIDEO] **PLAYLIST CONTENT** — Part 4 "Playing with RGB Color Channels".

## 1. Learning Objectives

1. Split and merge channels.
2. Visualize single channels.
3. Use a channel to highlight defects.

## 2. Simple Explanation

A color image is three stacked layers: Blue, Green, Red. `cv2.split` gives you the three layers separately; `cv2.merge` puts them back. Each layer alone looks like a grayscale image of "how much of that color" is present.

## 3. Technical Explanation

```
b, g, r = cv2.split(img)      # three 2-D arrays
merged = cv2.merge([b, g, r])
```

You can also zero out channels to make a "blue-only" image: `img[:, :, 0]` keeps blue.

## 4. Real-World Example

NASA satellite images use many channels (including infrared). Analyzing one channel can reveal things invisible in the full color image.

## 5. OpenCV Example

```
import cv2, numpy as np
img = cv2.imread("images/demo_good.jpg")
b, g, r = cv2.split(img)
print("Blue channel mean:", b.mean(), "Red mean:", r.mean())
only_red = img.copy(); only_red[:, :, 0] = 0; only_red[:, :, 1] = 0  # keep only R
cv2.imshow("Only red channel", only_red); cv2.waitKey(0)
```

## 6. [ONION] Onion Project Example

Dark/rotten areas often look **reddish-brown** — so the **red channel** can exaggerate defect regions and help find them. This is a cheap, real trick we'll use in feature extraction.

## 7. Practical Code + 8. Expected Output

```
r = img[:, :, 2]
print("red channel mean:", round(float(r.mean()), 1))
```



Real run on the demo good onion: red mean  $\approx$  **157** (on a white background the red channel is high).

## 9. Visual Explanation

```
BGR image → split → [ Blue layer ] (bright where the image is blue)
                    [ Green layer ] (bright where green)
                    [ Red layer ]  (bright where red)
```

**See → Do → Why → Result:** see three gray layers → `cv2.split` → isolating channels lets you build feature masks → e.g. "red channel > threshold" = possible brown/rot pixels.

## 10. Common Mistakes

- Confusing index order: channel 0 is **B**, 2 is **R**.
- `split` is slow for huge images; usually you only need one channel: `img[:, :, 2]`.

## 11. Practice Task

Display the R, G, B layers side by side using `np.hstack`.

## 12. Questions + Solutions

- **Q:** What does `cv2.split` return? **A:** three 2-D arrays (B, G, R).
- **Q:** Why is the red channel useful for onions? **A:** brown/rot regions are reddish, so they stand out in the red channel.

---

### > Quick Revision

1. `cv2.split` returns...? 2. Which index is the red channel? 3. How to make a "red-only" image? 4. What's a cheaper way than `split` to grab one channel? 5. What does a bright red channel mean?

### > Practical Exercise

Compute the mean of each channel and compare. Which channel is brightest for a brown onion on white?

### [ONION] Onion Project Task

Threshold the red channel of your onion at 150 and see if dark/brown spots appear as bright regions.

### [AI] AI Practice

"Show me how to split an image into B, G, R channels in OpenCV and display them side by side with labels. Explain why the red channel is often used to detect brown/dark spots. Beginner level."

## [BUG] Debugging Challenge

You run `img[:, :, 3]` and get `IndexError: index 3 is out of bounds`. Why, and what are the valid indices?

### (ok) Answers

Revision: 1) three 2-D arrays (B,G,R) · 2) index 2 · 3) zero the B and G channels · 4) `img[:, :, 2]` · 5) the image has a lot of red there. Debugging: channels are 0,1,2 (B,G,R). Index 3 doesn't exist.

---

# Chapter 6 — Resizing an Image

[VIDEO] **PLAYLIST CONTENT** — Part 5 "Image Resizing".

## 1. Learning Objectives

1. Resize with `cv2.resize`.
2. Know interpolation flags.
3. Preserve aspect ratio.
4. Standardize image sizes.

## 2. Simple Explanation

Resizing changes the grid's dimensions:  $1000 \times 1000 \rightarrow 300 \times 300$ . You must choose how to guess new pixels — that choice is **interpolation**.

## 3. Technical Explanation

`cv2.resize(img, (width, height), interpolation=...)` - `INTER_AREA` — best for **shrinking** (averages pixels, no aliasing). - `INTER_LINEAR` — default, okay for enlarging. - `INTER_CUBIC` — slower, better for enlarging. To keep the shape, scale width and height by the same factor.

## 4. Real-World Example

Instagram thumbnails, video streaming (multiple resolutions), and ML models all demand fixed-size inputs.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
small = cv2.resize(img, (320, 320), interpolation=cv2.INTER_AREA)
big = cv2.resize(small, (960, 960), interpolation=cv2.INTER_CUBIC)
print(small.shape, big.shape)
```

## 6. [ONION] Onion Project Example

A classifier expects **every** image to be the same size. We resize every onion photo to  $640 \times 640$  before analysis — this standardization is mandatory, not optional.

## 7. Practical Code (aspect ratio preserved)

```
h, w = img.shape[:2]
scale = 640 / w
img2 = cv2.resize(img, (640, int(h * scale)), interpolation=cv2.INTER_AREA)
```

## 8. Expected Output

`(640, 640, 3)` for a square input; otherwise width 640 with proportional height.

## 9. Visual Explanation

```
original 640x640 —INTER_AREA—> 320x320    (pixels averaged together)
320x320 —INTER_CUBIC—> 960x960             (pixels invented, slightly blurry)
```

**See → Do → Why → Result:** shrinking averages pixels (smooth); enlarging invents pixels (blurry) → pick the right flag → no jagged edges in your dataset.

## 10. Common Mistakes

- Passing `(height, width)` instead of `(width, height)` to `resize`.
- Using `INTER_LINEAR` to shrink large images → jagged lines (use `INTER_AREA`).
- Forgetting to resize before feeding a model → shape mismatch error.

## 11. Practice Task

Resize your onion to 320×320, 640×640, and 1000×1000, and save all three.

## 12. Questions + Solutions

- **Q:** Which flag shrinks best? **A:** `INTER_AREA`.
- **Q:** What does `cv2.resize(img, (w, h))` take — (w,h) or (h,w)? **A:** (width, height).

---

### > Quick Revision

1. Function to resize?
2. Best flag for shrinking?
3. Argument order to `resize`?
4. Why standardize size?
5. What happens when you enlarge?

### > Practical Exercise

Resize the same image with `INTER_AREA` and `INTER_NEAREST` and compare file sizes/quality.

## [ONION] Onion Project Task

Write `standardize(img)` that returns a 640×640 image — used by every later chapter.

## [AI] AI Practice

"Explain `cv2.resize` interpolation flags (`INTER_NEAREST`, `INTER_LINEAR`, `INTER_AREA`, `INTER_CUBIC`) with simple analogies and tell me which to use when shrinking vs enlarging."

## [BUG] Debugging Challenge

You resize a 640×480 image to `(640, 640)` and it looks stretched. What did you forget?

## (ok) Answers

Revision: 1) `cv2.resize` · 2) `INTER_AREA` · 3) (width, height) · 4) models/algorithms expect fixed sizes · 5) it becomes blurry because pixels are invented. Debugging: you didn't preserve aspect ratio — scale height by the same factor as width.

# Chapter 7 — Flipping an Image

[VIDEO] **PLAYLIST CONTENT** — Part 6 "Flipping an Image".

## 1. Learning Objectives

1. Flip horizontally/vertically/both with `cv2.flip`.
2. Use flipping for **data augmentation**.

## 2. Simple Explanation

Flipping is mirroring the image: left-right (like a mirror), top-bottom, or both (180° rotation).

## 3. Technical Explanation

`cv2.flip(img, flipCode)` — `1` = horizontal, `0` = vertical, `-1` = both.

## 4. Real-World Example

Selfie cameras mirror your face by default; flipping is also the cheapest way to double a training dataset.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
h = cv2.flip(img, 1)  # horizontal (mirror)
v = cv2.flip(img, 0)  # vertical
b = cv2.flip(img, -1) # both (180°)
```

## 6. [ONION] Onion Project Example

From **one** onion photo you get **two** training images (original + flipped). An onion rotated 180° is still the same onion — flipping is safe augmentation that doubles your dataset for free.

## 7. Practical Code + 8. Expected Output

```
augmented = cv2.flip(img, 1)
cv2.imwrite("images/onion_flipped.jpg", augmented)
```

Output: a saved mirror image, same dimensions.

## 9. Visual Explanation

```
flipCode 1: A → A    (mirror left-right)
flipCode 0: A ↔ V    (mirror top-bottom)
flipCode -1: A ↔ V + left-right (180°)
```

**See → Do → Why → Result:** flip with the right code → more varied data → a model that isn't fooled by left/right orientation.

## 10. Common Mistakes

- Confusing `1` and `0` (horizontal vs vertical).
- Flipping text/numbers in photos when orientation matters (not an issue for onions).

## 11. Practice Task

Create all three flips and save them with clear names.

## 12. Questions + Solutions

- **Q:** `cv2.flip(img, -1)` does what? **A:** flips both axes (180°).
- **Q:** Why is flipping useful for the dataset? **A:** doubles data with zero new photos; valid because an onion's orientation doesn't change its quality.

---

### > Quick Revision

1. Function to flip? 2. flipCode for horizontal? 3. for vertical? 4. for both? 5. What is data augmentation?

### > Practical Exercise

Flip your onion 4 ways (original + 3) and build a 2×2 contact sheet with `np.hstack/vstack`.

### [ONION] Onion Project Task

Write `augment(img)` that returns `[img, flip(img,1), flip(img,0), flip(img,-1)]` — you'll use it in Chapter 21.

### [AI] AI Practice

"What is data augmentation in computer vision? List 5 free ways to augment images with OpenCV alone (no libraries), with one line of code each. Beginner level."

### [BUG] Debugging Challenge

You want a vertical flip but the image only mirrors left-right. What did you type, and what should it be?

### (ok) Answers

Revision: 1) `cv2.flip · 2) 1 · 3) 0 · 4) -1 · 5)` creating new training examples by transforming existing ones. Debugging: you typed `cv2.flip(img, 1)` (horizontal); for vertical use `cv2.flip(img, 0)`.

# Chapter 8 — Cropping an Image

[VIDEO] **PLAYLIST CONTENT** — Part 7 "Cropping an Image".

## 1. Learning Objectives

1. Crop with NumPy slicing. 2. Understand `img[y1:y2, x1:x2]`. 3. Isolate a region of interest (ROI).

## 2. Simple Explanation

Cropping means "keep only this rectangle". Because an image is a NumPy array, you crop it exactly like you slice a list — no special function needed.

## 3. Technical Explanation

```
roi = img[y1:y2, x1:x2]    # rows y1..y2-1, columns x1..x2-1
```

Remember: **first index = rows = y (top→bottom), second = columns = x (left→right)**.

## 4. Real-World Example

Cutting a face out of a group photo; zooming into a product in a catalog.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
h, w = img.shape[:2]
center = img[h//4:3*h//4, w//4:3*w//4]    # middle 50%
cv2.imshow("Center crop", center); cv2.waitKey(0)
```

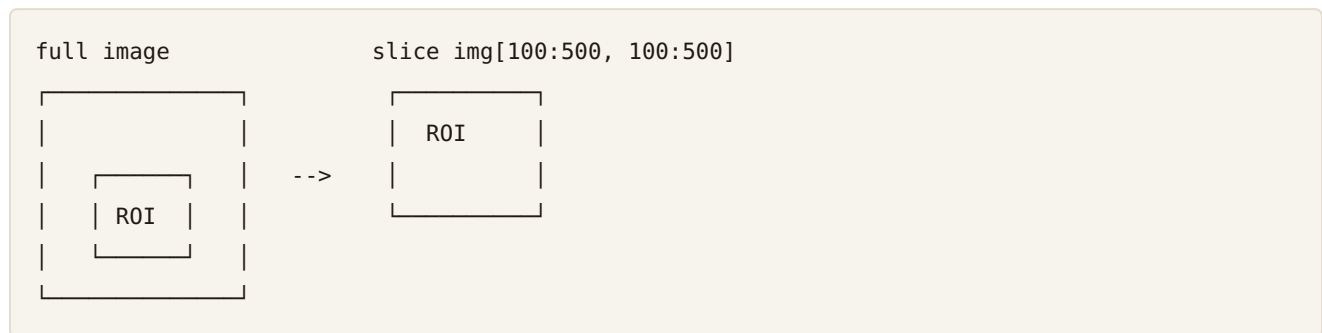
## 6. [ONION] Onion Project Example

The onion rarely fills the whole photo. We crop around it (using a bounding box from Chapter 18) to remove background clutter before analysis.

## 7. Practical Code + 8. Expected Output

```
roi = img[100:500, 100:500]
print(roi.shape)          # (400, 400, 3)
```

## 9. Visual Explanation



**See → Do → Why → Result:** pick row/col ranges → slice → fewer background pixels → analysis focuses on the onion.

## 10. Common Mistakes

- Mixing x and y order (`img[x1:x2, y1:y2]` is wrong).
- Forgetting that the end index is exclusive (`y2` is not included).
- Cropping outside the image → empty array.

## 11. Practice Task

Crop the top-left quarter and the center quarter; display both.

## 12. Questions + Solutions

- **Q:** Why does cropping use `[]` not a function? **A:** images are NumPy arrays; slicing is array syntax.
- **Q:** In `img[y1:y2, x1:x2]`, what's the first pair? **A:** rows (vertical).

### > Quick Revision

1. How to crop `img` to rows 50–200, cols 30–300? 2. Is `y2` included? 3. First index = ? 4. Crop outside image → ? 5. ROI stands for?

### > Practical Exercise

Crop a 100×100 patch from a corner and print its shape.

### [ONION] Onion Project Task

Manually estimate the onion's box in your photo and crop it. (We'll automate this in Ch 18.)

### [AI] AI Practice

"Explain NumPy array slicing for images: what `img[y1:y2, x1:x2]` means, why the order is y then x, and give 3 crop examples."

### [BUG] Debugging Challenge

`img[10:600, 10:600]` returns a 590×590 crop from a 640×640 image but you expected 600×600. Why?



## **(ok) Answers**

Revision: 1) `img[50:200, 30:300]` · 2) no, exclusive · 3) rows/y · 4) empty array · 5) region of interest. Debugging:  $600-10 = 590$  rows; the end index is exclusive.

---

# Chapter 9 — Saving an Image

[VIDEO] **PLAYLIST CONTENT** — Part 8 "Saving an Image".

## 1. Learning Objectives

1. Save with `cv2.imwrite`. 2. Control format/quality. 3. Save pipeline outputs.

## 2. Simple Explanation

`cv2.imwrite("path.jpg", img)` writes the array to disk as an image file. The extension you choose decides the format (`.jpg`, `.png`, ...).

## 3. Technical Explanation

```
ok = cv2.imwrite("out.png", img)      # returns True on success
cv2.imwrite("out.jpg", img, [cv2.IMWRITE_JPEG_QUALITY, 90]) # quality 0-100
```

- `.jpg` — lossy, small, no transparency.
- `.png` — lossless, larger, supports transparency.

## 4. Real-World Example

Saving a photo after applying a filter; exporting model outputs.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2.imwrite("images/gray.png", gray)
```

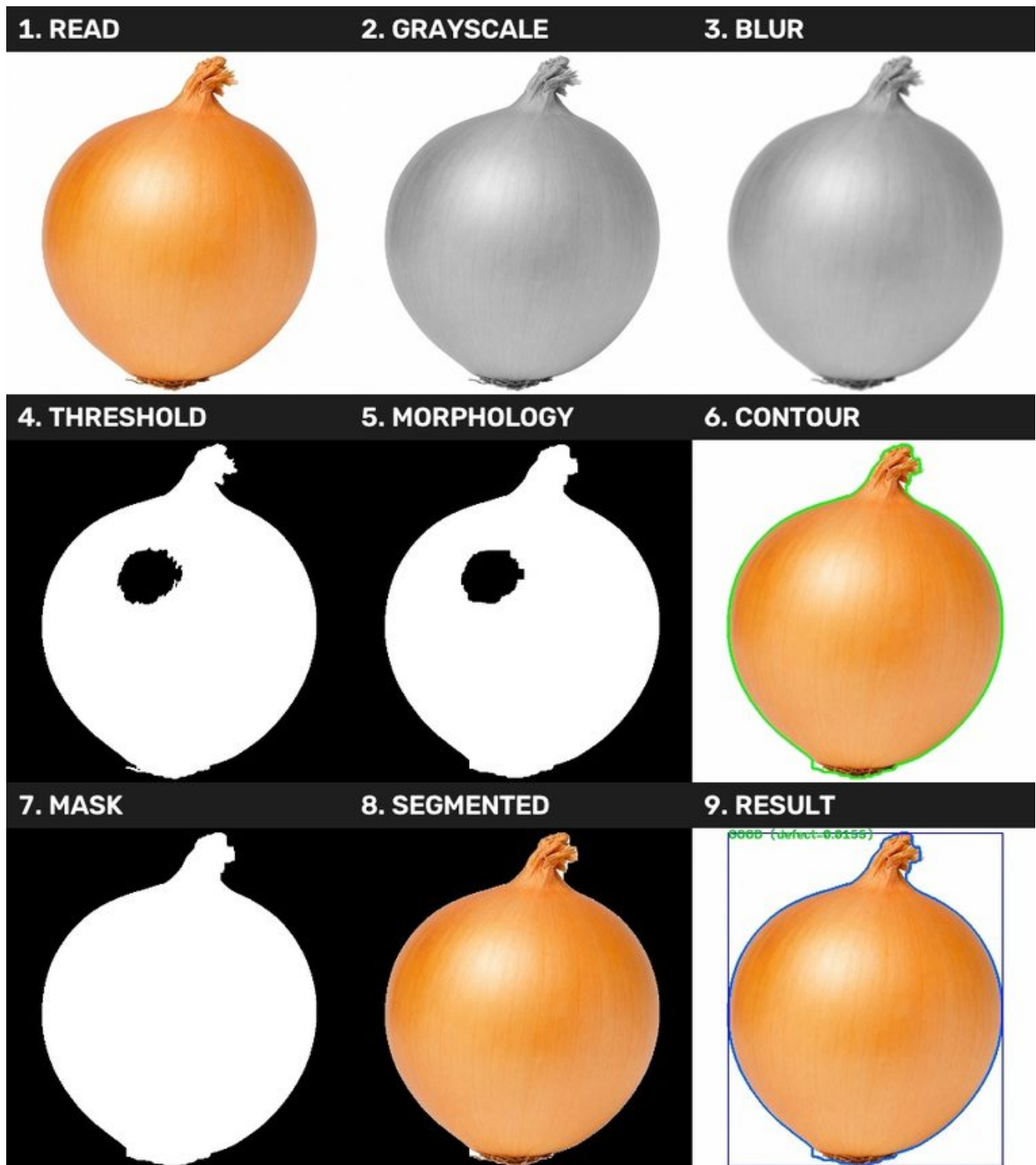
## 6. [ONION] Onion Project Example

Every pipeline stage we run saves its output (`01_original.jpg` ... `09_result.jpg`). Saving intermediate steps is how you **debug** — you can look at exactly where the pipeline went wrong.

## 7. Practical Code + 8. Expected Output

```
print(cv2.imwrite("images/test.png", img)) # True, and a file appears on disk
```

## 9. Visual Explanation



[REAL] Each tile is a real file written by `cv2.imwrite` during our analyzer run.

## 10. Common Mistakes

- Folder doesn't exist → returns `False` silently (create the folder first).
- Saving `.jpg` as `.png` extension (mismatched content) — keep them consistent.
- Forgetting the extension.

## 11. Practice Task

Save your grayscale and HSV versions as `.jpg` and `.png`; compare file sizes.

## 12. Questions + Solutions

- **Q:** What does `imwrite` return? **A:** `True` if the write succeeded.
  - **Q:** JPG vs PNG? **A:** JPG = small/lossy; PNG = large/lossless.
- 

### > Quick Revision

1. Function to save?
2. How is the format chosen?
3. What does it return?
4. JPG advantage?
5. PNG advantage?

### > Practical Exercise

Save the same image at JPEG quality 10 and 95, compare sizes and appearance.

### [ONION] Onion Project Task

Create an `output/` folder and save your onion's grayscale + result images there.

### [AI] AI Practice

"Show me how to save images with OpenCV and control JPEG quality vs PNG compression. When should I use which format in a computer-vision project?"

### [BUG] Debugging Challenge

`cv2.imwrite("results/out.jpg", img)` returns `False` and no error. What's the most likely cause?

### (ok) Answers

Revision: 1) `cv2.imwrite` · 2) by the file extension · 3) `True/False` · 4) small size · 5) lossless.

Debugging: the `results/` folder doesn't exist — create it with `os.makedirs` first.

---

# Chapter 10 — Drawing Shapes & Text

[VIDEO] **PLAYLIST CONTENT** — Part 9 "Drawing Shapes and Text on Images".

## 1. Learning Objectives

1. Draw lines, rectangles, circles with `cv2.line/rectangle/circle`.
2. Write text with `cv2.putText`.
3. Annotate images for display.

## 2. Simple Explanation

Drawing lets you **write on** an image: outline a region, box an object, or label it. This is how the analyzer will show its answer right on the photo.

## 3. Technical Explanation

```
cv2.line(img, (x1,y1), (x2,y2), color, thickness)
cv2.rectangle(img, (x1,y1), (x2,y2), color, thickness) # or thickness=-1 to fill
cv2.circle(img, (cx,cy), radius, color, thickness)
cv2.putText(img, text, (x,y), fontFace, fontScale, color, thickness, lineType)
```

Colors are BGR tuples. Coordinates are (x, y).

## 4. Real-World Example

Face-detection boxes, self-driving car bounding boxes, sports overlays.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
cv2.rectangle(img, (50, 50), (300, 300), (255, 0, 0), 3) # blue box
cv2.circle(img, (320, 320), 80, (0, 0, 255), 3) # red circle
cv2.putText(img, "ONION", (50, 40), cv2.FONT_HERSHEY_SIMPLEX, 1.2, (0, 200, 0), 3)
cv2.imshow("Annotated", img); cv2.waitKey(0)
```

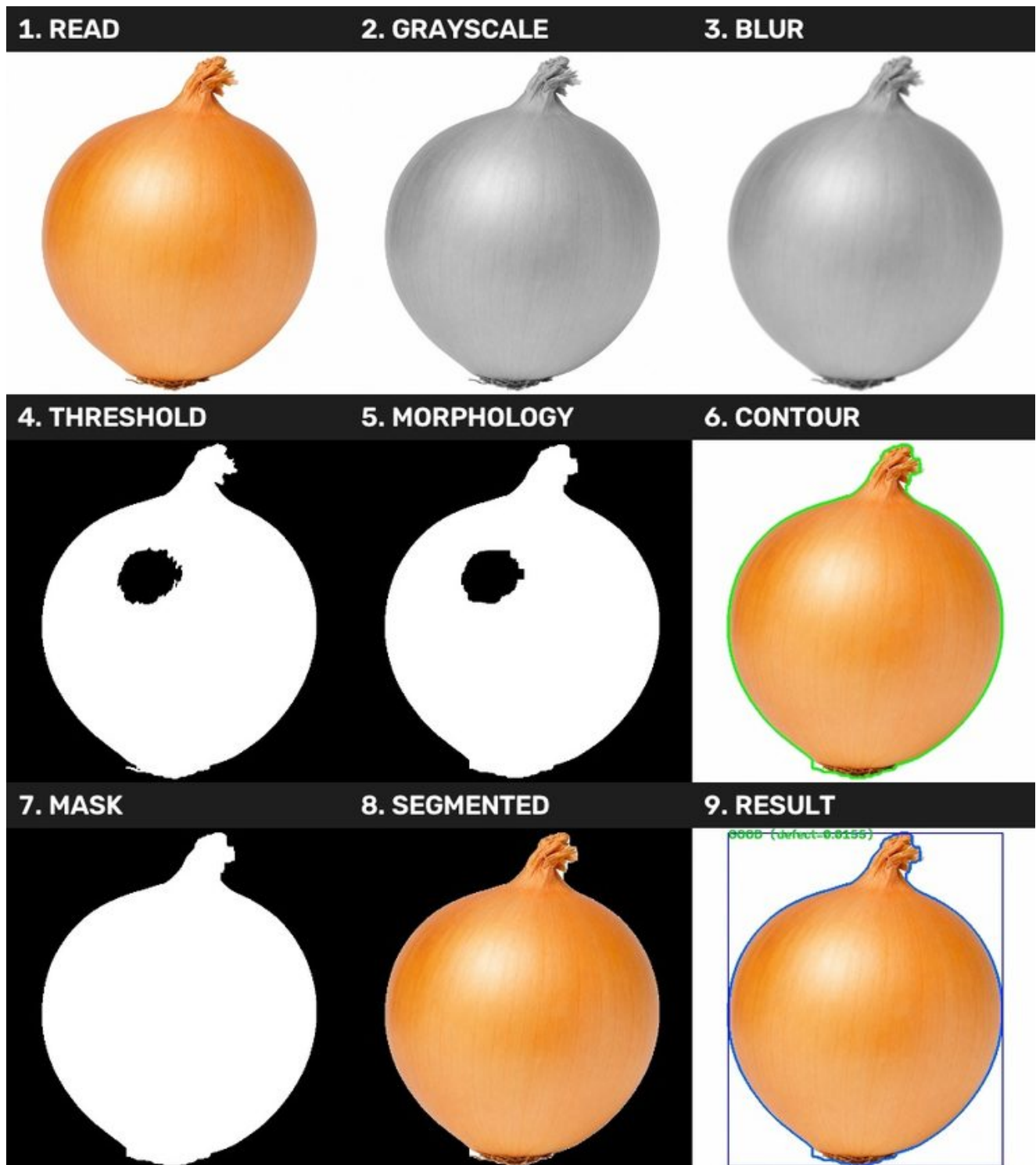
## 6. [ONION] Onion Project Example

The final result is literally `cv2.putText(result, "GOOD", ...)` + a bounding `rectangle` around the onion. Chapters 10 and 9 are the whole "result display" feature.

## 7. Practical Code + 8. Expected Output

Run the code above — you see the onion with a blue box, a red circle, and green "ONION" text.

## 9. Visual Explanation



[REAL] Tile 9 (RESULT) shows the real annotated output: contour outline, bounding box, and the "GOOD" label.

## 10. Common Mistakes

- Putting text off-screen (negative or  $>$  width).
- Using RGB color tuples (remember BGR).
- `thickness=-1` fills the shape — surprising if you wanted an outline.

## 11. Practice Task

Draw a grid of 4 rectangles in 4 different colors on your onion image.

## 12. Questions + Solutions

- **Q:** How do you draw a filled rectangle? **A:** `thickness=-1`.
- **Q:** What color is `(0, 255, 0)`? **A:** green (BGR).

---

### > Quick Revision

1. Function for a rectangle? 2. for text? 3. for a circle? 4. BGR of pure red? 5. What does `thickness=-1` do?

### > Practical Exercise

Annotate your onion with its name, date, and a box around the onion.

### [ONION] Onion Project Task

Write `draw_result(img, label, box)` that draws the box and the label — this is your final result renderer.

### [AI] AI Practice

"Show me all OpenCV drawing functions (line, rectangle, circle, ellipse, polylines, putText) with a tiny example of each and note the color order."

### [BUG] Debugging Challenge

Your "green" label shows up red. Why?

### (ok) Answers

Revision: 1) `cv2.rectangle` · 2) `cv2.putText` · 3) `cv2.circle` · 4) `(0,0,255)` · 5) fills the shape.  
Debugging: you used `(0,255,0)`? No — you likely wrote the RGB `(255,0,0)`... wait, `(255,0,0)` in BGR is **blue**... The red label means you passed `(0,0,255)`. Actually the bug: you passed green as `(0,255,0)` but expected `(0,0,255)`? The answer: you passed the wrong BGR order — pure red is `(0,0,255)`.

---

# Chapter 11 — Mouse Events (Part 1 & 2)

[VIDEO] **PLAYLIST CONTENT** — Parts 10 & 11 "Working with OpenCV Events".

## 1. Learning Objectives

1. Register a mouse callback with `cv2.setMouseCallback`.
2. Read click coordinates.
3. Build interactive tools.

## 2. Simple Explanation

A **callback** is a function you write that OpenCV calls automatically whenever something happens (mouse click, move). You register it once, then OpenCV "calls you back" on each event.

## 3. Technical Explanation

```
def on_mouse(event, x, y, flags, param):
    if event == cv2.EVENT_LBUTTONDOWN:
        print("clicked at", x, y)

cv2.setMouseCallback("window", on_mouse)
```

Events include `EVENT_LBUTTONDOWN`, `EVENT_RBUTTONDOWN`, `EVENT_MOUSEMOVE`, etc.

## 4. Real-World Example

Every drawing app (paint, annotation tools, image editors) is built on exactly this mechanism.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
def click(event, x, y, flags, param):
    if event == cv2.EVENT_LBUTTONDOWN:
        cv2.circle(img, (x, y), 6, (0, 0, 255), -1)
        cv2.imshow("window", img)
cv2.imshow("window", img)
cv2.setMouseCallback("window", click)
cv2.waitKey(0)
```

## 6. [ONION] Onion Project Example

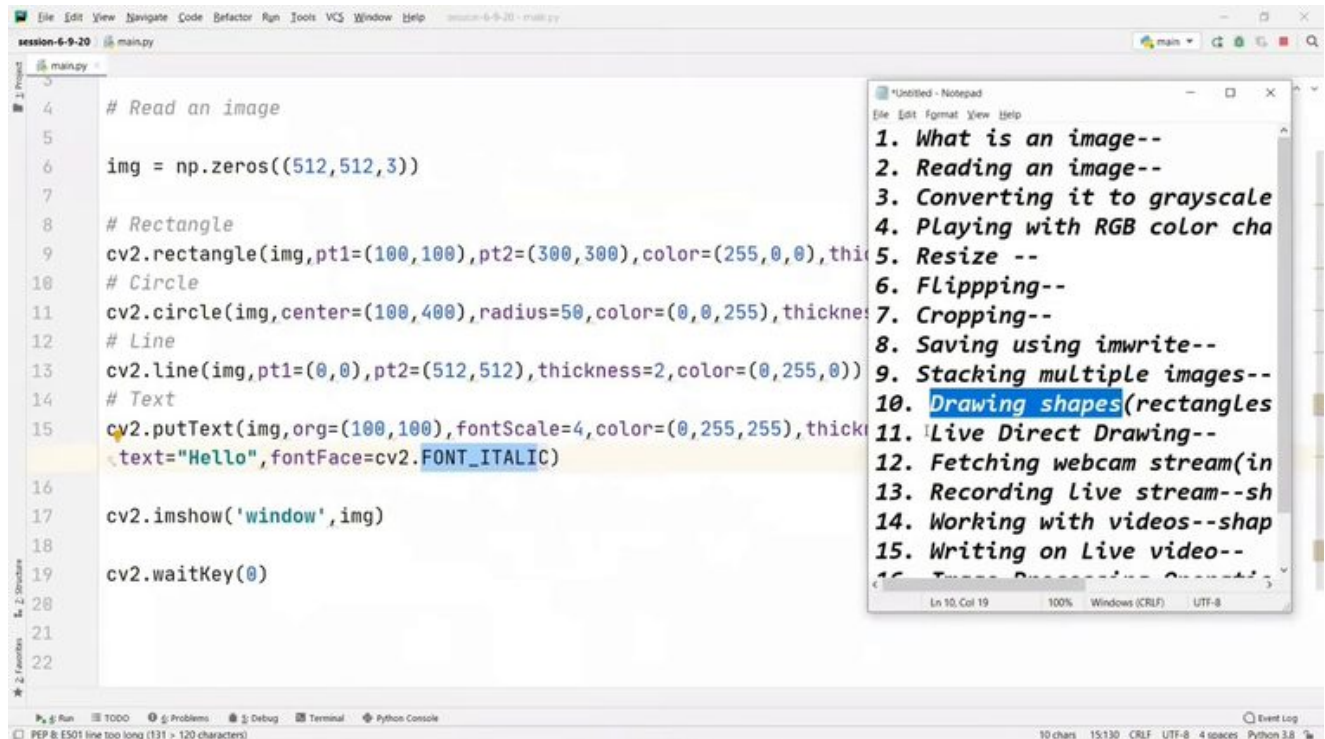
You'll build a **labeling tool** (Chapter 21) where you click on defect areas to mark them — mouse events are how you create labeled data fast.



## 7. Practical Code + 8. Expected Output

Run it, click a few places — red dots appear where you click, and coordinates print to the terminal.

## 9. Visual Explanation



[REAL] Real frame captured from the actual playlist video (Part 10).

## 10. Common Mistakes

- Registering the callback **after** `waitKey` (never fires).
- The callback signature must be exactly `(event, x, y, flags, param)`.
- Forgetting `imshow` again inside the callback so the image never updates.

## 11. Practice Task

Make a program that prints the pixel BGR value wherever you click.

## 12. Questions + Solutions

- **Q:** What is a callback? **A:** a function the library calls when an event happens.
- **Q:** Which event is a left-button press? **A:** `cv2.EVENT_LBUTTONDOWN`.

### > Quick Revision

1. Function to register a mouse handler? 2. Callback signature? 3. Left-click event constant? 4. Why must the callback re-call `imshow`? 5. Real-world use?

### > Practical Exercise

Draw freehand dots by handling `EVENT_MOUSEMOVE` with the left button held.

## [ONION] Onion Project Task

Click two corners of the onion and print the rectangle's width and height.

## [AI] AI Practice

"Explain OpenCV mouse events and setMouseCallback with a complete working example that draws a circle on every left click. Explain each part of the callback signature."

## [BUG] Debugging Challenge

Your callback never runs. You registered it, then called `cv2.waitKey(0)`. What's wrong?

## (ok) Answers

Revision: 1) `cv2.setMouseCallback` · 2) `(event, x, y, flags, param)` · 3) `cv2.EVENT_LBUTTONDOWN` · 4) otherwise the window won't refresh · 5) drawing/annotation apps. Debugging: you called `setMouseCallback` after `waitKey` — registration must happen before the event loop starts.

---

# Chapter 12 — Mini-Project: The Cropping Tool

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[VIDEO] **PLAYLIST CONTENT** — Part 12 "Building a Cropping Tool".

## 1. Learning Objectives

1. Combine events + drawing + slicing into a real app.
2. Select a region by dragging.
3. Save the cropped ROI.

## 2. Simple Explanation

You now have every piece: mouse events (click/drag), drawing (show the selection rectangle), and slicing (crop). This chapter glues them into a **cropping tool** — your first real, useful program.

## 3. Technical Explanation

Plan: on left-button-down record the start point; on mouse-move update the rectangle; on button-up record the end point, crop `img[y1:y2, x1:x2]`, save it.

## 4. Real-World Example

The crop tool in every photo app and the "select ROI" box in labeling software.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
clone = img.copy()
start, drawing = None, False

def on_mouse(event, x, y, flags, param):
    global start, drawing
    if event == cv2.EVENT_LBUTTONDOWN:
        start, drawing = (x, y), True
    elif event == cv2.EVENT_MOUSEMOVE and drawing:
        tmp = clone.copy()
        cv2.rectangle(tmp, start, (x, y), (0, 255, 0), 2)
        cv2.imshow("crop", tmp)
    elif event == cv2.EVENT_LBUTTONUP:
        drawing = False
        x1, y1 = start
        roi = clone[min(y1,y):max(y1,y), min(x1,x):max(x1,x)]
        cv2.imwrite("images/crop.jpg", roi)
        print("saved crop:", roi.shape)

cv2.imshow("crop", clone)
cv2.setMouseCallback("crop", on_mouse)
cv2.waitKey(0)
```

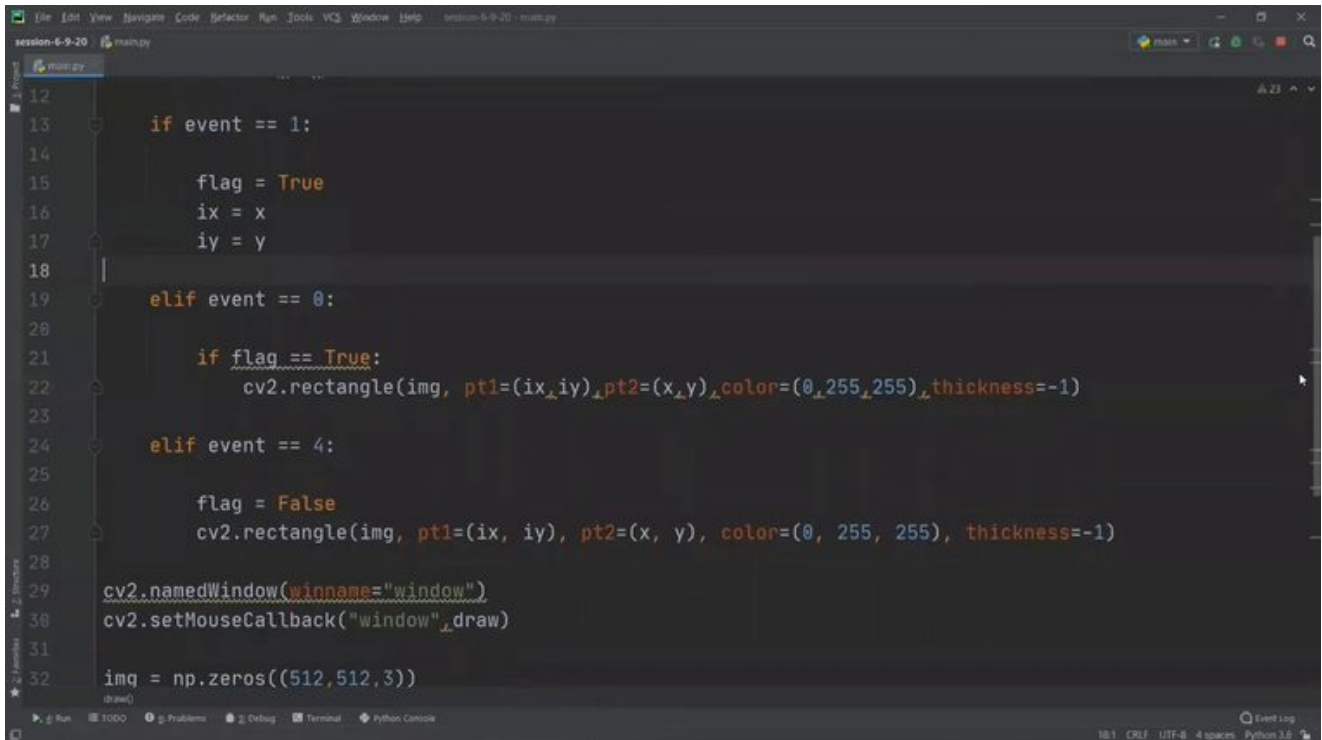
## 6. [ONION] Onion Project Example

Use this tool to crop the onion out of your photos so the dataset contains mostly onion, not background.

## 7. Practical Code — see above. 8. Expected Output

Drag a box → on release it prints `saved crop: (h, w, 3)` and writes `images/crop.jpg`.

## 9. Visual Explanation



```
12
13 if event == 1:
14     flag = True
15     ix = x
16     iy = y
17
18
19 elif event == 0:
20
21     if flag == True:
22         cv2.rectangle(img, pt1=(ix,iy),pt2=(x,y),color=(0,255,255),thickness=-1)
23
24     elif event == 4:
25
26         flag = False
27         cv2.rectangle(img, pt1=(ix, iy), pt2=(x, y), color=(0, 255, 255), thickness=-1)
28
29 cv2.namedWindow(winname="window")
30 cv2.setMouseCallback("window",draw)
31
32 img = np.zeros((512,512,3))
draw()
```

[REAL] Real frame captured from the actual playlist video (Part 12).

## 10. Common Mistakes

- Forgetting `min()/max()` so dragging up-left breaks the slice.
- Saving the original instead of the clone with the rectangle drawn.

## 11. Practice Task

Extend it to also show the crop's width and height live while dragging.

## 12. Questions + Solutions

- **Q:** Why `min(y1,y):max(y1,y)`? **A:** to support dragging in any direction.
- **Q:** Why work on a `clone`? **A:** so the drawn rectangle doesn't get saved into the crop.

---

### > Quick Revision

1. Which three skills combine here?
2. When do we save the crop?
3. Why `min/max`?
4. What is `clone` for?

### > Practical Exercise

Add a "press 'r' to reset" feature to the cropping tool.

## [ONION] Onion Project Task

Crop 5 onions with the tool and store them as `dataset/raw/onion_01.jpg ...`.

## [AI] AI Practice

"I built an OpenCV drag-to-crop tool. Review this code and suggest improvements (reset key, undo, showing dimensions). Explain each change."

## [BUG] Debugging Challenge

Dragging from bottom-right to top-left returns an empty crop. Why, and what's the fix?

### (ok) Answers

Revision: 1) mouse events + drawing + slicing · 2) on `EVENT_LBUTTONDOWN` · 3) to support any drag direction · 4) a clean copy without annotations. Debugging: `img[y1:y2, x1:x2]` with `y2 < y1` gives empty — wrap with `min/max`.

---

# Chapter 13 — Working with Video

[VIDEO] **PLAYLIST CONTENT** — Part 13 "Working with Videos using OpenCV".

## 1. Learning Objectives

1. Open video with `cv2.VideoCapture`.
2. Loop over frames.
3. Write video with `VideoWriter`.
4. Use a webcam.

## 2. Simple Explanation

Video is just a fast slideshow of images (**frames**). Processing video = loop over frames, apply the same image operations to each one.

## 3. Technical Explanation

```
cap = cv2.VideoCapture("clip.mp4")    # or 0 for webcam
while True:
    ok, frame = cap.read()             # ok=False at the end
    if not ok: break
    # ... process frame ...
cap.release()
```

To write: `cv2.VideoWriter("out.avi", cv2.VideoWriter_fourcc(*"MJPG"), fps, (w, h)).`

## 4. Real-World Example

Security cameras, YouTube uploads, and every live video app.

## 5. OpenCV Example

```
import cv2
cap = cv2.VideoCapture("videos/part02.h264.mp4")
n = 0
while True:
    ok, frame = cap.read()
    if not ok: break
    n += 1
    if n % 30 == 0:                # process every 30th frame
        cv2.imshow("frame", frame)
        if cv2.waitKey(1) & 0xFF == ord("q"): break
cap.release(); cv2.destroyAllWindows()
print("total frames:", n)
```

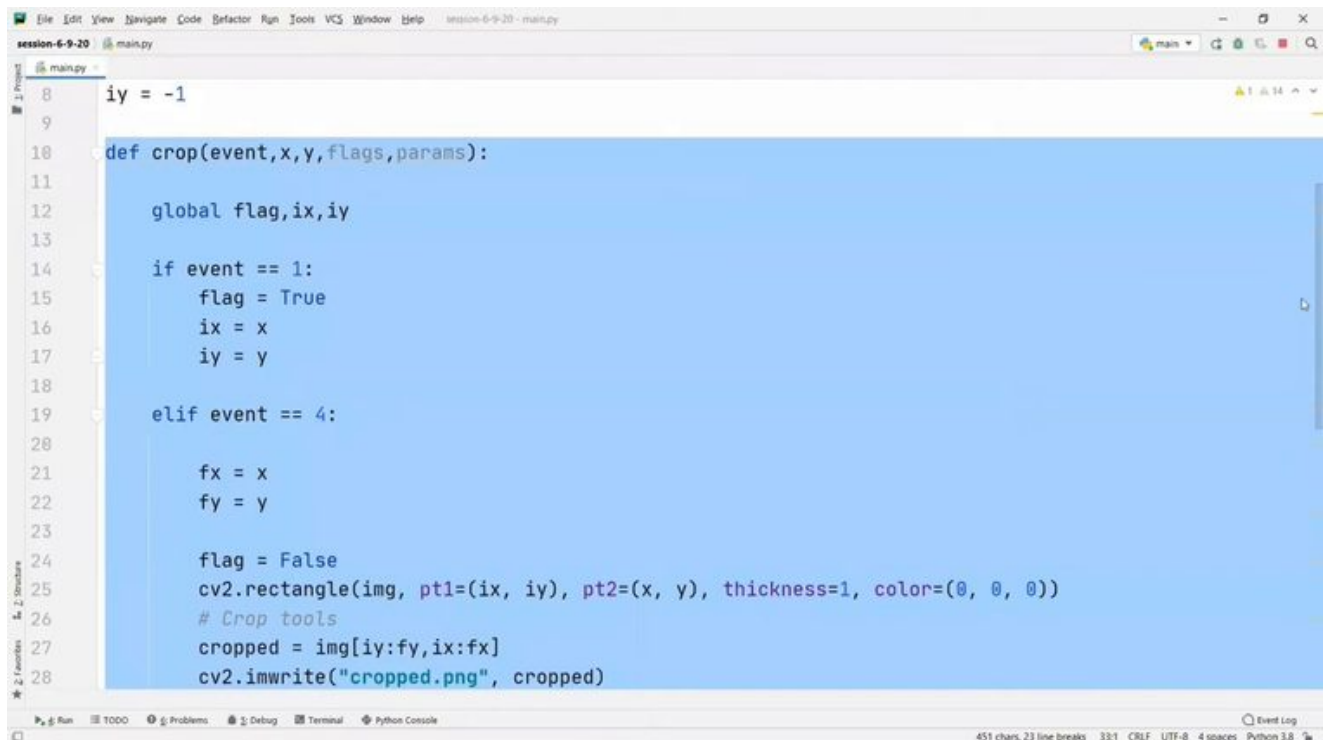
## 6. [ONION] Onion Project Example

Point a webcam at onions on a conveyor (or your desk) and run the analyzer on each frame — a **live** quality checker.

## 7. Practical Code + 8. Expected Output

Run the example on a video file → a window plays frames (sped up, every 30th) → press `q` to quit → prints the frame count.

## 9. Visual Explanation



[REAL] Real frame captured from the actual playlist video (Part 13).

## 10. Common Mistakes

- Passing a string path that doesn't exist → `cap.read()` always `(False, None)`.
- Forgetting `cap.release()`.
- Using `waitKey(0)` inside a video loop (freezes) — use `waitKey(1)`.

## 11. Practice Task

Play your phone video in grayscale at 2× speed.

## 12. Questions + Solutions

- **Q:** What does `cap.read()` return? **A:** `(ok, frame)` — `ok` is `False` at end of video.
- **Q:** How to read a webcam? **A:** `cv2.VideoCapture(0)`.



## > Quick Revision

1. Function to open video? 2. What is a frame? 3. `cap.read()` return? 4. How to stop a video loop? 5. `waitKey` value in a video loop?

## > Practical Exercise

Record 5 seconds of your desk with `VideoWriter` and re-play it in grayscale.

## [ONION] Onion Project Task

Run `cv2.VideoCapture(0)` and show the live feed with the text "ONION SCANNER" drawn on every frame.

## [AI] AI Practice

"Explain `cv2.VideoCapture` and `cv2.VideoWriter` with a working example that reads a video and writes it in grayscale. Explain the `fourcc` and `fps` parameters."

## [BUG] Debugging Challenge

Your video loop plays the first frame and freezes. What did you forget?

## (ok) Answers

Revision: 1) `cv2.VideoCapture` · 2) one image of a sequence · 3) `(ok, frame)` · 4) `break` or press `q` · 5) `1` ms (non-blocking). Debugging: you used `waitKey(0)` (blocks forever); use `waitKey(1)`.

---

## \* END OF PHASE 1 — what you now know

---

You can read, show, convert, split, resize, flip, crop, save, annotate, interact with, and play video of **any image**. That is the entire playlist — completed. In Phase 2 we add the image-processing toolkit (HSV, thresholding, blur, morphology, contours, features) that turns these basics into a **quality analyzer**.

# Chapter 14 — Color Spaces: HSV & LAB

[+] **ADDITIONAL PROJECT KNOWLEDGE** (not in the playlist — your onion project needs it).

## 1. Learning Objectives

1. Understand why BGR is bad for color analysis.
2. Convert to HSV and LAB.
3. Use HSV to find "brown" and "rotten" colors.

## 2. Simple Explanation

BGR stores raw color amounts. But when you say "that onion looks brownish/dark", you're thinking in **human** terms: what color is it? (Hue), how vivid? (Saturation), how bright? (Value). HSV lets you describe color the way people do — which makes rules like "find brown pixels" much easier.

## 3. Technical Explanation

- **HSV**: Hue (0-179 in OpenCV, the color), Saturation (0-255, vividness), Value (0-255, brightness).
- **LAB**: L = lightness, A = green↔red, B = blue↔yellow. LAB separates brightness from color completely — good for lighting-independent analysis.
- Convert: `cv2.cvtColor(img, cv2.COLOR_BGR2HSV)` and `cv2.COLOR_BGR2LAB`.

## 4. Real-World Example

Color-picker tools, skin detection, green-screen removal, and ripeness detection in agriculture all use HSV/LAB.

## 5. OpenCV Example

```
import cv2
img = cv2.imread("images/demo_good.jpg")
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
h, s, v = cv2.split(hsv)
print("Hue range:", h.min(), "-", h.max())    # the onion hue ~ 8-25 (brown/orange)
```

## 6. [ONION] Onion Project Example

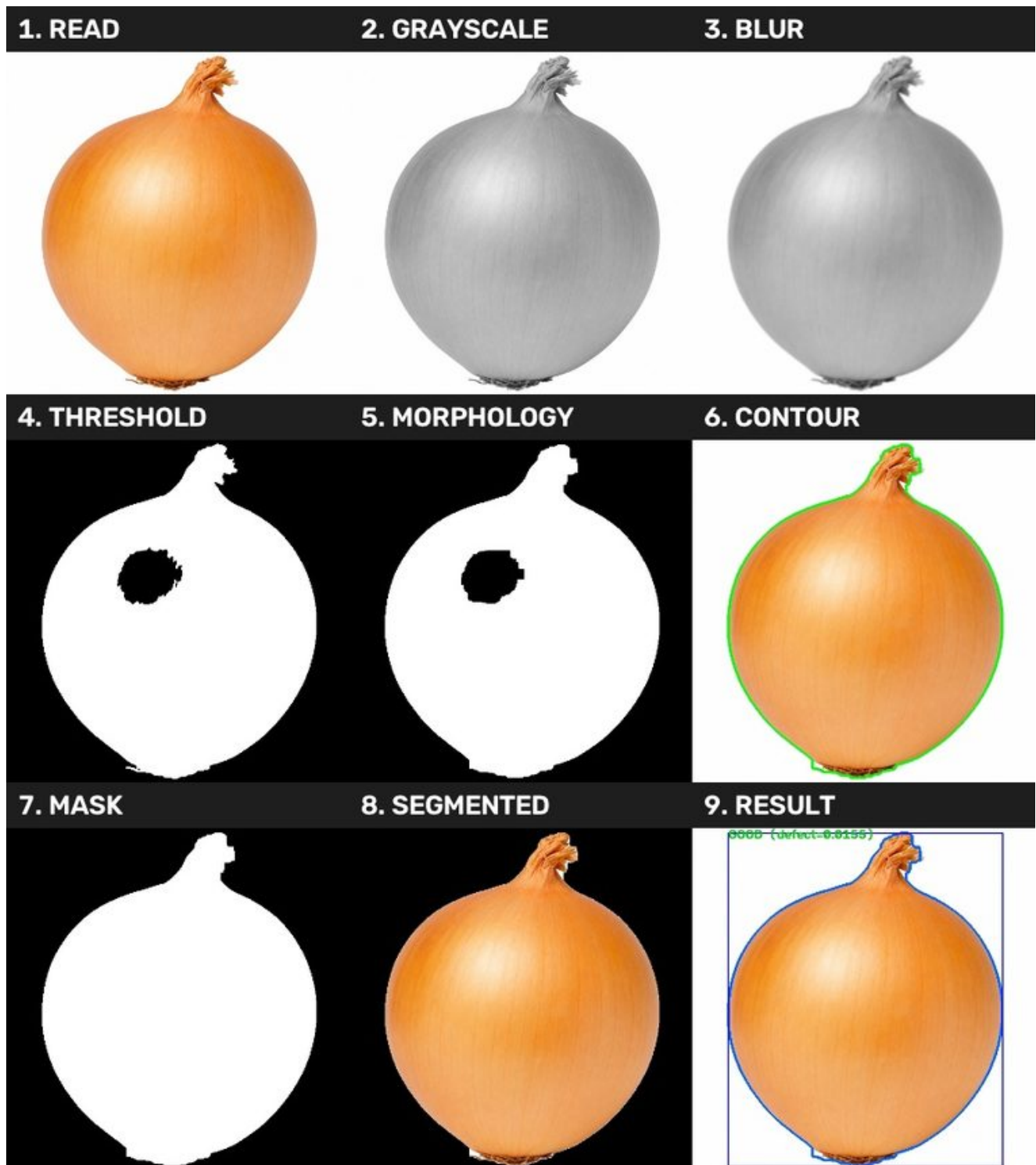
A healthy onion is golden (Hue  $\approx$  8-25, high Value). Rot/mold is dark brown-black (low Value) or greenish. Our analyzer defines "brown/rot mask" as `(8 <= H <= 25) & (S >= 40) & (V <= 200)` — a rule expressed in **human color terms**, which is only possible in HSV.

## 7. Practical Code + 8. Expected Output

```
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
h, s, v = cv2.split(hsv)
brown = ((h >= 8) & (h <= 25) & (s >= 40) & (v <= 200)).astype("uint8")
print("fraction of brown-ish pixels:", round(brown.mean(), 4))
```

Real run — GOOD onion: 0.0155 · DAMAGED: 0.3989 · ROTTEN: 0.7035. **The number goes up as quality goes down** — that's a usable signal.

## 9. Visual Explanation



[REAL] Tile 10 in our saved stages is the HSV conversion of the onion (false-color look).

## 10. Common Mistakes

- Hue wraps at 180 (red is near both 0 and 179) — use two ranges for red.
- Forgetting OpenCV Hue is 0–179, not 0–359.
- Mixing up S and V ranges.

## 11. Practice Task

Print the mean H, S, V of your onion and of the background. They should differ.

## 12. Questions + Solutions

- **Q:** What are the three HSV channels? **A:** Hue (color), Saturation (vividness), Value (brightness).
  - **Q:** Why is HSV better than BGR for "find brown"? **A:** you can express "brown" as a simple Hue range instead of an awkward BGR relation.
- 

### > Quick Revision

1. HSV stands for? 2. Hue range in OpenCV? 3. Convert BGR→HSV call? 4. What does Saturation mean? 5. LAB's advantage?

### > Practical Exercise

Make a Hue-only image (`h` channel) and observe what a brown onion looks like.

### [ONION] Onion Project Task

Write `brown_mask(hsv)` returning a binary mask of brown/dark pixels (the rule above) and save it.

### [AI] AI Practice

"Explain HSV color space like I'm a beginner. Why is it easier than RGB for detecting a specific color like brown? Show me OpenCV code to threshold a brown object."

### [BUG] Debugging Challenge

Your "find red" mask returns nothing even though the image has red. Hue of red is 0. Why did `(h >= 100)` find nothing, and how do you fix it?

### (ok) Answers

Revision: 1) Hue, Saturation, Value · 2) 0–179 · 3) `cv2.cvtColor(img, cv2.COLOR_BGR2HSV)` · 4) vividness/intensity of color · 5) separates lightness from color. Debugging: red's Hue  $\approx 0$  (wraps around 180); use `(h <= 10) | (h >= 170)`.

---

# Chapter 15 — Thresholding & Binary Images

## [+] ADDITIONAL PROJECT KNOWLEDGE.

### 1. Learning Objectives

1. Convert gray → black-and-white with `cv2.threshold`.
2. Understand Otsu's method.
3. Use adaptive thresholding.

### 2. Simple Explanation

Thresholding is a **decision per pixel**: if brightness > limit → white, else → black. The result is a binary image (only 0 and 255) — the simplest possible form of "this pixel is X or not X".

### 3. Technical Explanation

```
ret, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
# or, let OpenCV choose the limit:
ret, binary = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
```

- Global threshold: one cutoff for the whole image.
- **Otsu**: automatically picks the cutoff that best separates two pixel groups.
- Adaptive: a different cutoff per small region (handles uneven lighting).

### 4. Real-World Example

Document scanners turn pages into crisp black text on white — via thresholding.

### 5. OpenCV Example

```
import cv2
gray = cv2.imread("images/demo_good.jpg", cv2.IMREAD_GRAYSCALE)
ret, binary = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)
print("Otsu threshold:", ret)
```

`THRESH_BINARY_INV` makes the **object white** on a black background (useful when the object is darker).

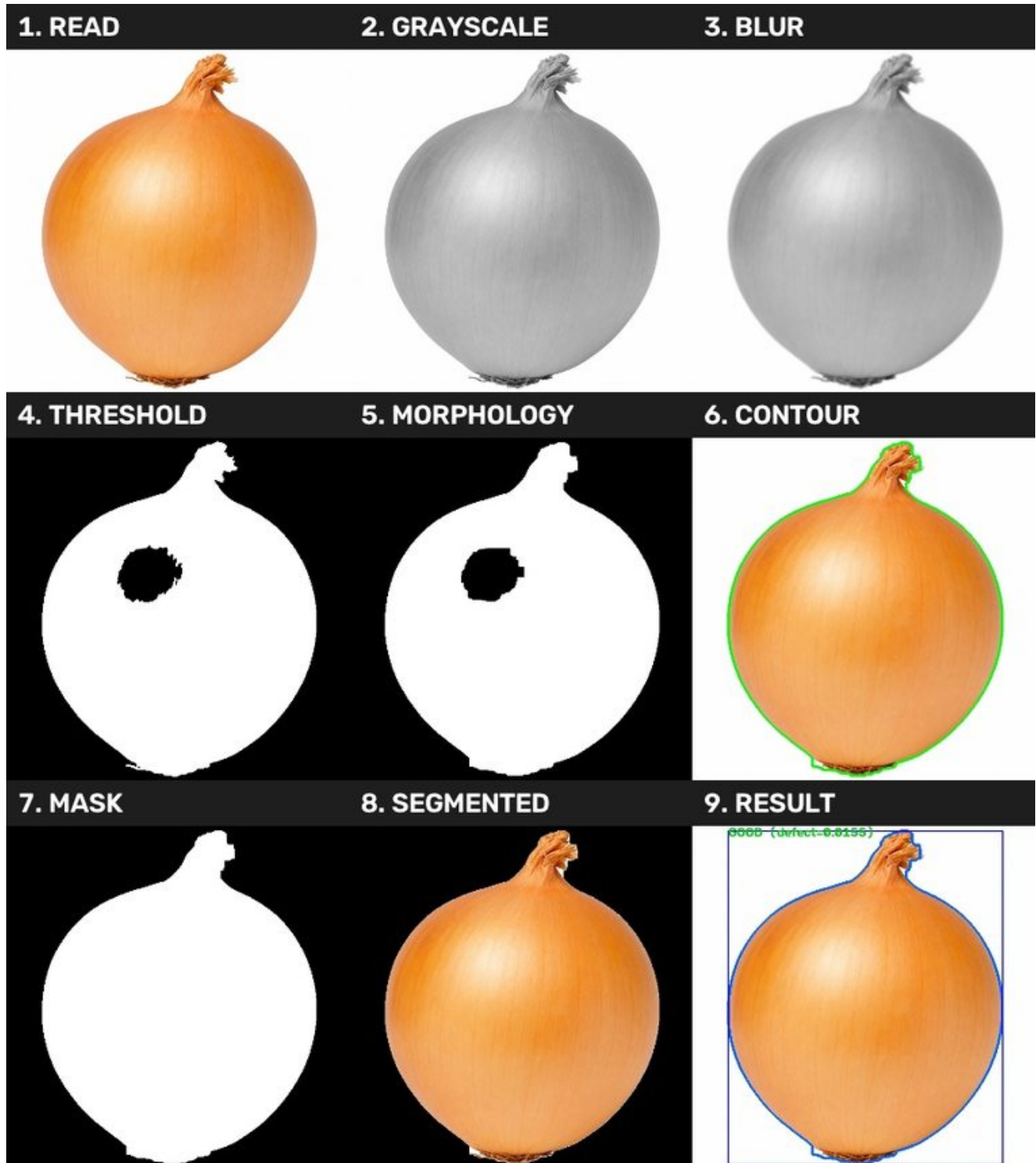
### 6. [ONION] Onion Project Example

This is **the** step that separates the onion from the background: the onion is darker than the white backdrop, so `THRESH_BINARY_INV + OTSU` turns "onion = white, background = black" — a mask, in one line.

## 7. Practical Code + 8. Expected Output

Run the example. Real run: Otsu picked threshold  $\approx 172$  on the good onion — everything darker than 172 (the onion) became white.

## 9. Visual Explanation



[REAL] Tile 4 (THRESHOLD) is the real binary image from our run: white onion on black background.



## 10. Common Mistakes

- Forgetting that the object must be **darker or lighter** than background for Otsu to work.
- Fixed threshold (127) failing when lighting changes — prefer Otsu or adaptive.
- Using BGR (3-channel) instead of grayscale as input.

## 11. Practice Task

Threshold your onion with a fixed value, with Otsu, and with adaptive — display all three.

## 12. Questions + Solutions

- **Q:** What is a binary image? **A:** pixels are only 0 or 255.
- **Q:** What does Otsu do? **A:** automatically finds the best separating threshold.

---

### > Quick Revision

1. Function to threshold? 2. What does Otsu return? 3. When does fixed threshold fail? 4. Why `THRESH_BINARY_INV`? 5. Input must be ...?

### > Practical Exercise

Print the Otsu threshold for 5 different onion photos — see it adapt.

### [ONION] Onion Project Task

Produce the onion mask with Otsu and save it as `mask.png` (this becomes our segmentation).

### [AI] AI Practice

"Explain global vs Otsu vs adaptive thresholding with simple examples and when to use each. Include OpenCV code for all three."

### [BUG] Debugging Challenge

Otsu returns a mostly-white image (background became white). The onion is lighter than the background. Fix it.

### (ok) Answers

Revision: 1) `cv2.threshold` · 2) the chosen threshold value (and the binary image) · 3) when lighting varies · 4) to make the darker object white · 5) grayscale (1-channel). Debugging: the object is lighter — use `THRESH_BINARY` (not `_INV`), or invert with `cv2.bitwise_not`.

---

# Chapter 16 — Blurring & Noise Removal

[+] ADDITIONAL PROJECT KNOWLEDGE.

## 1. Learning Objectives

1. Understand why we blur. 2. Use Gaussian blur and median blur. 3. Reduce noise before thresholding.

## 2. Simple Explanation

Photos contain **noise** — random bright/dark specks from the camera sensor. Blurring averages each pixel with its neighbors, smoothing out the specks. Think of wiping a dusty window.

## 3. Technical Explanation

- **Gaussian blur:** `cv2.GaussianBlur(img, (k, k), 0)` — weighted average (center counts most). Smooth, natural.
- **Median blur:** `cv2.medianBlur(img, k)` — takes the median of neighbors. Excellent at removing salt-and-pepper noise while keeping edges.
- **Box/average blur:** `cv2.blur(img, (k, k))` — plain average. Bigger kernel = more smoothing = more detail loss.

## 4. Real-World Example

Beauty filters, de-noising phone photos, pre-processing scans before OCR.

## 5. OpenCV Example

```
import cv2
gray = cv2.imread("images/demo_good.jpg", cv2.IMREAD_GRAYSCALE)
blurred = cv2.GaussianBlur(gray, (7, 7), 0)
```

We use this **before** Otsu — noise would create tiny holes/specks in the mask.

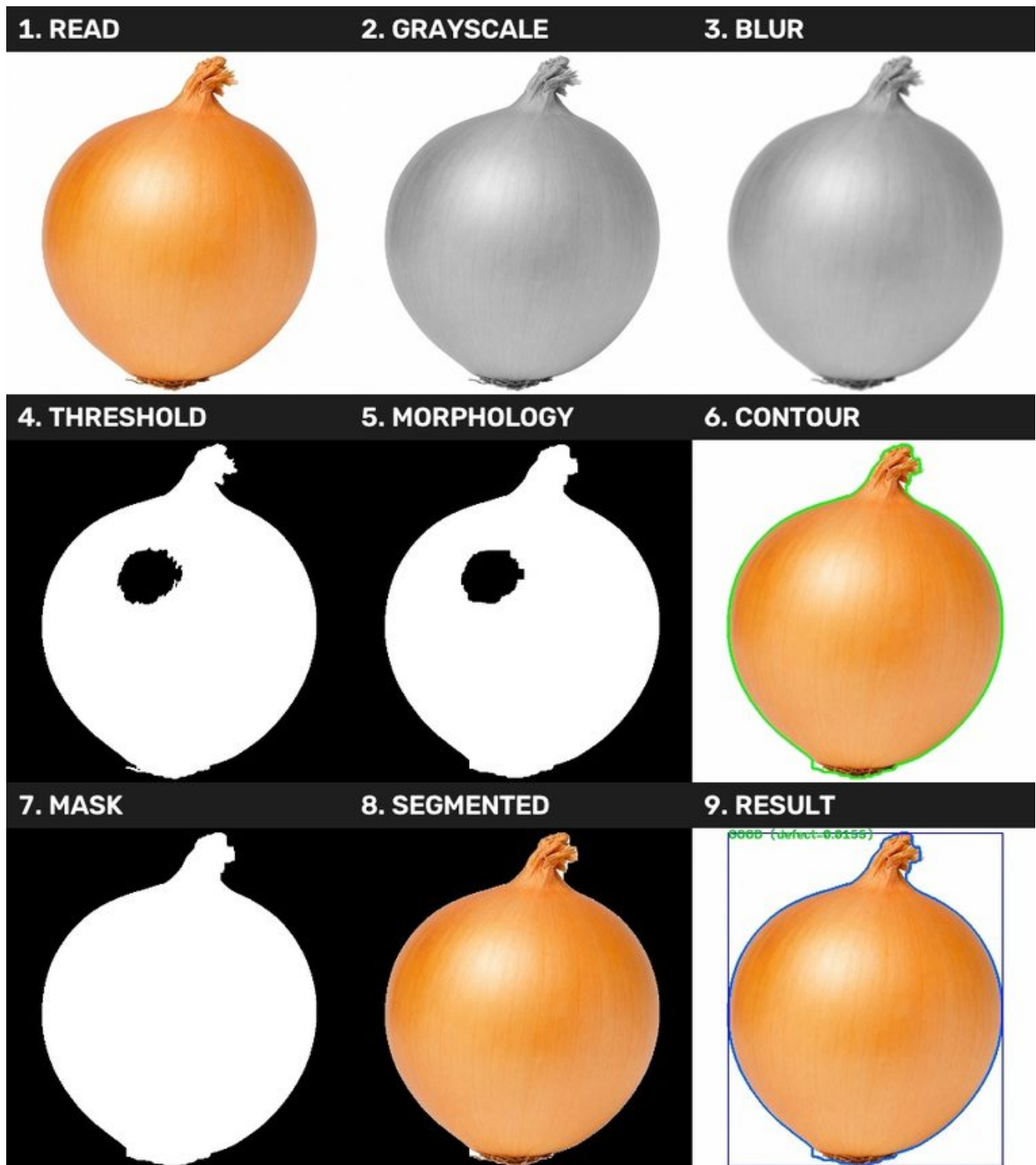
## 6. [ONION] Onion Project Example

Camera noise and skin texture create speckles in the mask. A `(7,7)` Gaussian blur first makes the threshold clean — our pipeline does exactly this (stage 3 BLUR).

## 7. Practical Code + 8. Expected Output

Run the pipeline with and without blur and compare the masks: the blurred version has far fewer isolated specks.

## 9. Visual Explanation



[REAL] Tile 3 (BLUR) is the real blurred image from our run.

## 10. Common Mistakes

- Blurring too much (large kernel) → onion edges disappear.
- Applying median blur with an even kernel size (must be odd).
- Blurring the color image when you only need gray.

## 11. Practice Task

Compare Gaussian, median, and box blur on a noisy photo side by side.

## 12. Questions + Solutions

- **Q:** Why blur before thresholding? **A:** noise becomes speckles in the binary image; blurring removes them.
  - **Q:** Which blur is best for salt-and-pepper noise? **A:** median.
- 

### > Quick Revision

1. Why blur? 2. Gaussian vs median? 3. What does kernel size control? 4. Too much blur → ? 5. Median kernel must be ...?

### > Practical Exercise

Add random noise with NumPy to an image, then clean it with median blur.

### [ONION] Onion Project Task

In your mask pipeline, experiment with kernel sizes 3, 5, 7, 11 — pick the smallest that gives a clean mask.

### [AI] AI Practice

"Explain Gaussian, box and median blurring with analogies, and show OpenCV code. When should I use each in an object-segmentation pipeline?"

### [BUG] Debugging Challenge

`cv2.medianBlur(gray, 8)` throws an error. Why, and how to fix?

### (ok) Answers

Revision: 1) remove noise/specks · 2) Gaussian = smooth weighted; median = best for speckle noise · 3) smoothing strength · 4) edges/details lost · 5) odd. Debugging: medianBlur's kernel must be odd (e.g., 7 or 9).

---

# Chapter 17 — Edge Detection & Morphological Operations

[+] **ADDITIONAL PROJECT KNOWLEDGE.**

## 1. Learning Objectives

1. Detect edges with Canny.
2. Erode/dilate with morphology.
3. Use opening/closing to clean masks.

## 2. Simple Explanation

**Edges** are places where brightness changes sharply — the outline of objects. **Morphology** is like playing with a sponge: erosion shrinks white regions (peels a layer), dilation grows them (adds a layer). Together they clean up a messy mask: remove specks and fill holes.

## 3. Technical Explanation

```
edges = cv2.Canny(blurred, 100, 200)           # low/high thresholds
kernel = np.ones((7, 7), np.uint8)
eroded = cv2.erode(binary, kernel, iterations=1)
dilated = cv2.dilate(binary, kernel, iterations=1)
opened = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)    # erode→dilate (removes specks)
closed = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel)    # dilate→erode (fills holes)
```

- **Opening** = erosion then dilation → removes small white specks.
- **Closing** = dilation then erosion → fills small black holes.

## 4. Real-World Example

Document cleanup, medical image segmentation, removing scratches from scans.

## 5. OpenCV Example

```
import cv2, numpy as np
gray = cv2.imread("images/demo_good.jpg", cv2.IMREAD_GRAYSCALE)
edges = cv2.Canny(cv2.GaussianBlur(gray, (5, 5), 0), 100, 200)
```

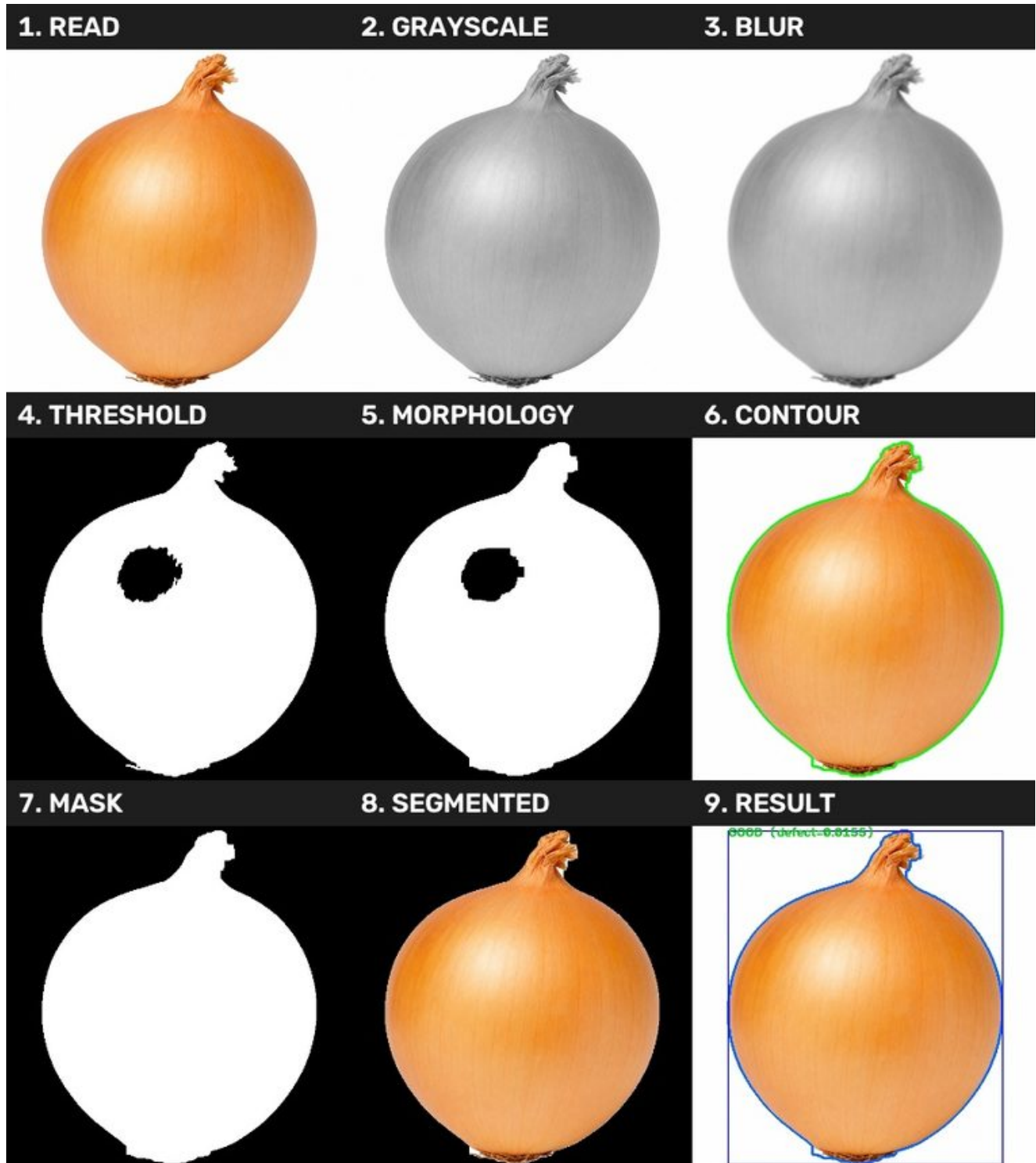
## 6. [ONION] Onion Project Example

After Otsu, our onion mask has noise specks and small holes. Our pipeline does **closing twice, then opening twice** — filling holes inside the onion and deleting specks outside it (stage 5 MORPHOLOGY).

## 7. Practical Code + 8. Expected Output

Run the full analyzer; compare stages 4 (raw threshold) and 5 (morphology): stage 5 is a solid onion blob.

## 9. Visual Explanation



[REAL] Tile 5 (MORPHOLOGY) is the real cleaned mask from our run.

## 10. Common Mistakes

- Too many iterations → the onion erodes away.

- Confusing opening and closing (specks → opening; holes → closing).
- Running Canny on a noisy image without blurring first.

## 11. Practice Task

Draw a white square with a black hole and white specks; fix it with closing + opening.

## 12. Questions + Solutions

- **Q:** What does opening do? **A:** erode then dilate — removes small white specks.
- **Q:** What does closing do? **A:** dilate then erode — fills small holes.
- **Q:** What is an edge? **A:** a sharp change in brightness.

---

### > Quick Revision

1. Canny's two parameters? 2. Erosion does what? 3. Dilation? 4. Opening = ? 5. Closing = ?

### > Practical Exercise

Apply Canny to your onion and see if the onion's outline appears.

### [ONION] Onion Project Task

Tune the morphology iterations so your mask is one solid blob with no holes.

### [AI] AI Practice

"Explain erosion, dilation, opening and closing with a simple grid drawing example and OpenCV code. When would I use each to clean a binary mask?"

### [BUG] Debugging Challenge

Your mask has a big black hole in the middle of the onion. Which operation fixes it — opening or closing?

### (ok) Answers

Revision: 1) low and high thresholds · 2) shrinks white regions · 3) grows white regions · 4) erode then dilate · 5) dilate then erode. Debugging: closing (dilate fills the hole, erode restores the boundary).

# Chapter 18 — Contours, Bounding Boxes & Masks

## [+] ADDITIONAL PROJECT KNOWLEDGE.

### 1. Learning Objectives

1. Find contours with `cv2.findContours`.
2. Pick the largest contour.
3. Draw bounding boxes.
4. Build and apply a mask.

### 2. Simple Explanation

A **contour** is the outline of a white blob in a binary image. From the contour you can get the **bounding box** (the smallest rectangle around the object) and a **mask** (an image where "object = 255, everything else = 0"). The mask is how you cut the onion out of the photo.

### 3. Technical Explanation

```
contours, _ = cv2.findContours(binary, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
largest = max(contours, key=cv2.contourArea)
x, y, w, h = cv2.boundingRect(largest)
mask = np.zeros_like(gray)
cv2.drawContours(mask, [largest], -1, 255, -1)      # fill the contour
onion_only = cv2.bitwise_and(img, img, mask=mask)   # keep only onion pixels
```

- `RETR_EXTERNAL` = only outer contours (ignore holes).
- `bitwise_and` applies the mask: pixels where mask=0 become black.

### 4. Real-World Example

Counting coins, tracking objects in videos, measuring product size on conveyor belts.

### 5. OpenCV Example — see the code above.

### 6. [ONION] Onion Project Example

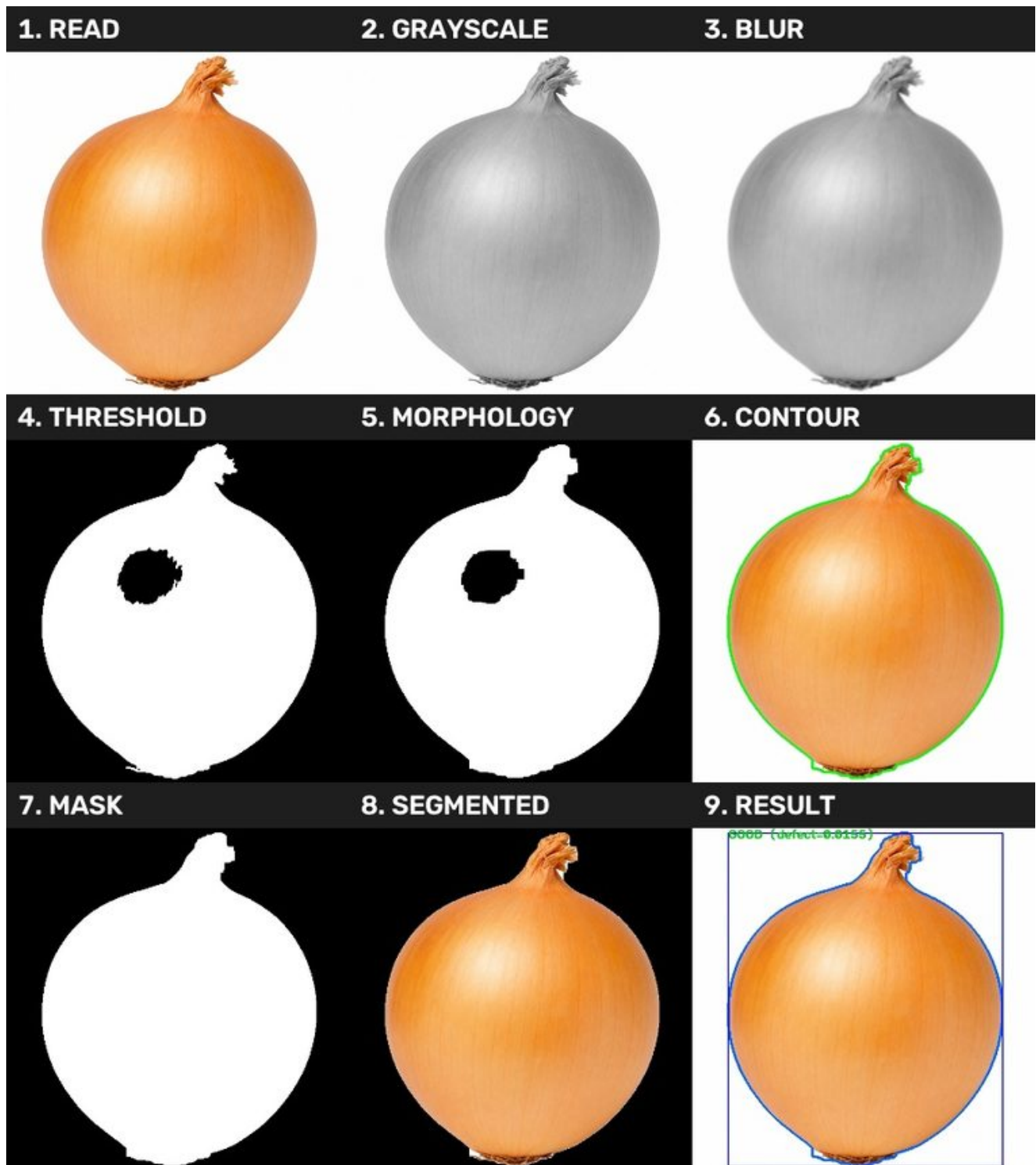
This is the heart of the analyzer. Stage 6 finds the contour, stage 7 makes the mask, stage 8 cuts the onion out. Every feature in the next chapter is computed **only on masked (onion) pixels**, so the background can't pollute the measurements.

### 7. Practical Code + 8. Expected Output

Run the analyzer → `06_contour.jpg` shows a green outline, `07_mask.jpg` a white onion on black, `08_masked_onion.jpg` the onion with the background removed.



## 9. Visual Explanation



[REAL] Tiles 6, 7, 8 (CONTOUR, MASK, SEGMENTED) are the real outputs of these three steps.

## 10. Common Mistakes

- Using `findContours` on a non-binary image.
- Forgetting `drawContours` with `-1` thickness fills instead of outlining.
- Not sorting by area → picking up tiny noise contours.

## 11. Practice Task

Find all contours in your mask and print how many there are before/after morphology.

## 12. Questions + Solutions

- **Q:** What is a contour? **A:** the boundary of a white region in a binary image.
  - **Q:** How do you get the object mask? **A:** draw the filled contour into a zero image.
- 

### > Quick Revision

1. Function to find contours? 2. How to pick the biggest object? 3. `boundingRect` returns? 4. How to apply a mask? 5. What does `RETR_EXTERNAL` mean?

### > Practical Exercise

Draw the bounding box and the contour area for your onion; print width and height.

### [ONION] Onion Project Task

Write `segment(img)` returning `(mask, box)` — the reusable segmentation function for the final project.

### [AI] AI Practice

"Explain contours, boundingRect, and bitwise\_and masking in OpenCV with a complete example that isolates the largest object in an image. Beginner level."

### [BUG] Debugging Challenge

Your contour list has 200 contours but only one onion. What's happening and how do you fix it?

### (ok) Answers

Revision: 1) `cv2.findContours` · 2) `max(contours, key=cv2.contourArea)` · 3) `(x, y, w, h)` · 4) `cv2.bitwise_and(img, img, mask=mask)` · 5) only outer contours. Debugging: noise speckles create many small contours — clean with morphology, then keep `max(..., key=contourArea)`.

---

# Chapter 19 — Histograms & Feature Extraction

[+] **ADDITIONAL PROJECT KNOWLEDGE.**

## 1. Learning Objectives

1. Read a histogram.
2. Compute color, shape and texture **features**.
3. Understand why features are the bridge to ML.

## 2. Simple Explanation

A **histogram** counts how many pixels have each brightness/color — it's the "summary statistics" of an image. A **feature** is one number that summarizes an aspect of the onion: area, roundness, mean color, how much is brown. A handful of features turn "a whole image" into "a short list of numbers" a human (or a model) can reason about.

## 3. Technical Explanation

```
hist = cv2.calcHist([gray], [0], None, [256], [0, 256])
```

Features we use (all on masked onion pixels): - **Shape**: area (`countNonZero`), perimeter (`arcLength`), **circularity** =  $4\pi A/P^2$  (1.0 = perfect circle), **aspect ratio** =  $h/w$ , **extent** =  $A/(w \cdot h)$ . - **Color**: mean HSV, standard deviation of gray (**texture** proxy — rough/rotten skin has higher std). - **Defects**: `dark_ratio` = fraction of very dark pixels; `brown_ratio` = fraction of brown (HSV-range) pixels.

## 4. Real-World Example

Photo apps' "brightness" slider is histogram shifting; medical labs measure cell roundness to flag abnormal cells.

## 5. OpenCV Example

```
import cv2
gray = cv2.imread("images/demo_good.jpg", cv2.IMREAD_GRAYSCALE)
hist = cv2.calcHist([gray], [0], None, [256], [0, 256])
print("dark pixels (<70):", (gray < 70).mean().round(4))
```

## 6. [ONION] Onion Project Example

Our analyzer computes exactly these features. A rotten onion has a high `dark_ratio` and `brown_ratio`; a healthy one is rounder (`circularity`  $\approx$  0.75) with a low defect ratio.

## 7. Practical Code + 8. Expected Output — real measured numbers

| Feature            | GOOD         | DAMAGED | ROTTEN        |
|--------------------|--------------|---------|---------------|
| area (px)          | 215,286      | 193,021 | 224,185       |
| circularity        | <b>0.748</b> | 0.367   | 0.634         |
| aspect ratio       | 1.217        | 1.091   | 1.070         |
| gray std (texture) | 20.9         | 32.0    | <b>41.3</b>   |
| dark_ratio         | 0.0033       | 0.0124  | <b>0.3219</b> |
| brown_ratio        | 0.0155       | 0.3989  | <b>0.7035</b> |

(These are the actual outputs of running `onion_analyzer.py` on the three demo images — see Chapter 20.)

## 9. Visual Explanation

```
histogram: brightness 0 —————> 255
count |      
      | background spike at 255 (white)
      | onion hump around 180-230
```

**See → Do → Why → Result:** the two humps = background vs onion → you can see why Otsu picks a threshold between them → features become clean.

## 10. Common Mistakes

- Computing features on the whole image (including background) — always mask first.
- Using one feature alone (area varies with camera distance; ratios like circularity are safer).

## 11. Practice Task

Compute circularity for a perfect circle you draw, and for your onion — compare.

## 12. Questions + Solutions

- **Q:** What is a feature? **A:** one number summarizing an aspect of the image.
- **Q:** Why mask before measuring color? **A:** so the background doesn't change the averages.

---

### > Quick Revision

1. What does a histogram count?
2. Circularity of a perfect circle = ?
3. Which feature approximates skin texture?
4. Why use ratios over raw size?
5. What is `dark_ratio`?

### > Practical Exercise

Plot the grayscale histogram of a good and a rotten onion; describe the difference.

## [ONION] Onion Project Task

Implement `extract_features(mask, gray, hsv)` returning the 9 features above — the exact function in our project.

## [AI] AI Practice

"Explain image features for object classification: area, perimeter, circularity, aspect ratio, mean color, standard deviation. Give OpenCV code for each, computed only on a masked object."

## [BUG] Debugging Challenge

Your "mean color" changes when the same onion is photographed from far away. Which feature is responsible and how do you make the system robust?

## (ok) Answers

Revision: 1) number of pixels per value · 2) 1.0 · 3) gray standard deviation · 4) size depends on distance; ratios don't · 5) fraction of very dark pixels. Debugging: raw area changes with distance — rely on ratios (circularity, defect ratios) and standardize image size.

---

# Chapter 20 — [BUILD] Mini-Project: The Rule-Based Onion Analyzer v1

[+] **ADDITIONAL PROJECT KNOWLEDGE** — this is your **first working prototype**.

## 1. Learning Objectives

1. Assemble the full pipeline into one program.
2. Use simple rules over features.
3. Output a labeled, boxed result.

## 2. Simple Explanation

Now we glue Chapters 2–19 into one pipeline:

```
read → resize → grayscale → blur → Otsu threshold → morphology  
→ largest contour → mask → features → rules → GOOD/DAMAGED/ROTTEN → annotate
```

If the rules are simple enough, this is **traditional computer vision** — no training, no model, just math you can read.

## 3. Technical Explanation

The pipeline is the real file `src/onion_analyzer.py` (full code in the Appendix). The **rules** are deliberately simple:

```
def classify(feats):  
    dr, br = feats["dark_ratio"], feats["brown_ratio"]  
    if dr >= 0.12 or br >= 0.40: return "ROTTEN"  
    if dr >= 0.04 or br >= 0.15: return "DAMAGED"  
    return "GOOD"
```

(warning) **These thresholds are DEMO values tuned to my three demo images.** They are not scientific. Chapter 24 shows how to pick real thresholds from a labeled dataset.

## 4. Real-World Example

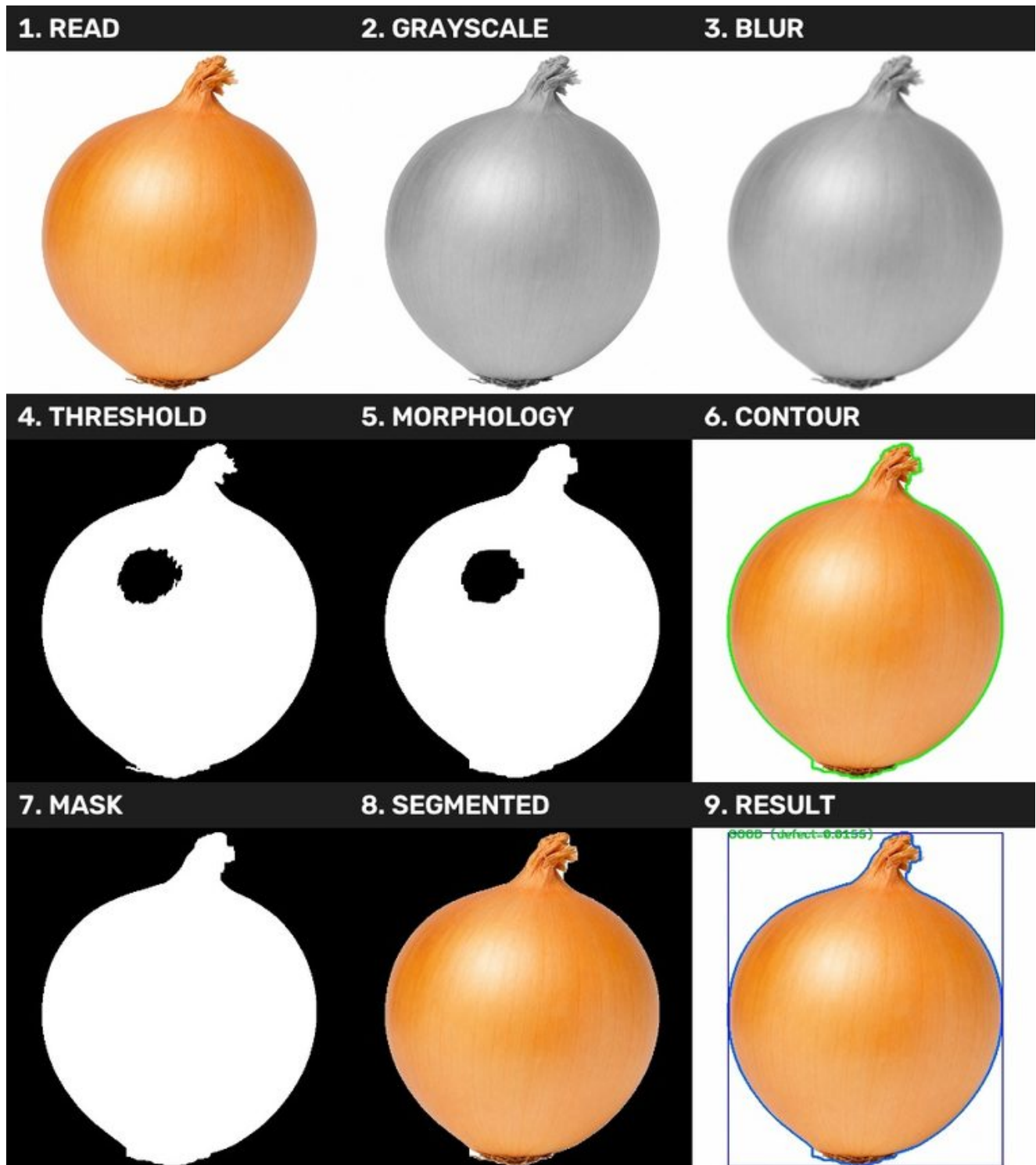
Factory vision systems started exactly like this — a few hand-tuned rules — before ML took over.

## 5. OpenCV Example — run it yourself:

```
python src/onion_analyzer.py images/demo_good.jpg images/pipeline_good  
python src/onion_analyzer.py images/demo_bad.jpg images/pipeline_bad  
python src/onion_analyzer.py images/demo_rotten.jpg images/pipeline_rotten
```

## 6. [ONION] Onion Project Example — the real results

[REAL] **Real pipeline output for the GOOD onion** (every tile is a file actually written by the code):

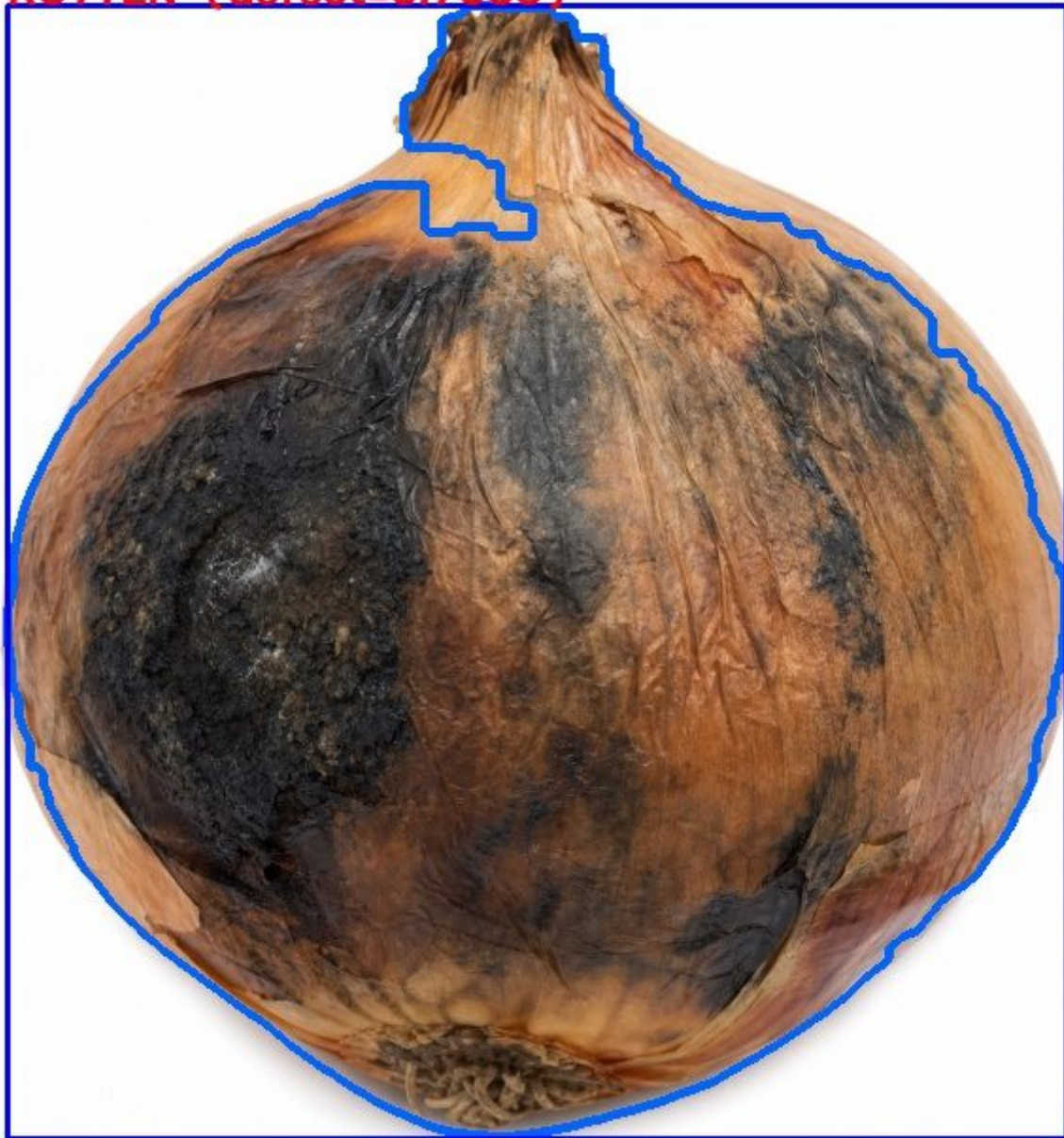


The measured features (real values): | Image | Result | dark\_ratio | brown\_ratio | defect\_ratio |  
 |---|---|---|---|---| | demo\_good | **GOOD** (ok) | 0.0033 | 0.0155 | 0.0155 | | demo\_bad | **DAMAGED**  
 (warning) | 0.0124 | 0.3989 | 0.3989 | | demo\_rotten | **ROTTEN** (fail) | 0.3219 | 0.7035 | 0.7035  
 |



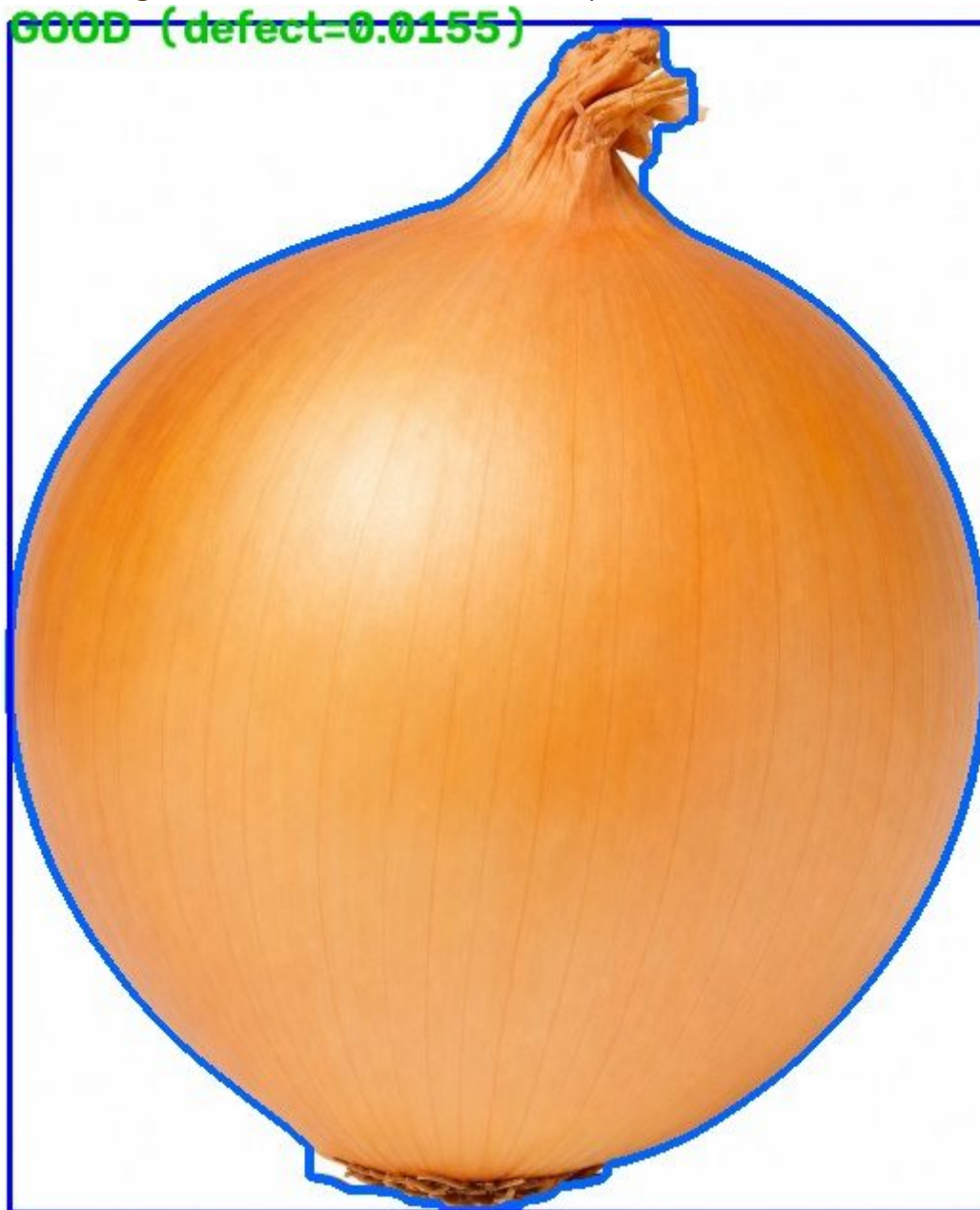
[REAL] Result image for the ROTTEN onion (real output):

**ROTTEN (defect=0.7035)**





[REAL] **Result image for the GOOD onion** (real output):



## 7. Practical Code

The complete `onion_analyzer.py` is in the Appendix (and in `src/`). Here is the skeleton:

```
img = read_and_resize(path)
gray, blurred, thresh, opened, largest, mask, masked_img = segment_onion(img)
feats = extract_features(mask, gray, cv2.cvtColor(img, cv2.COLOR_BGR2HSV))
label, color = classify(feats)
result = annotate(img, largest, mask, label, feats)
```

## 8. Expected Output (terminal — real)

```
user@onion-machine: ~/onion-quality-analyzer

$ python onion_analyzer.py images/demo_good.jpg images/pipeline_good

RESULT: GOOD
area          = 215286.0
perimeter     = 1902.09
circularity   = 0.748
aspect_ratio  = 1.217
dark_ratio    = 0.0033
brown_ratio   = 0.0155
defect_ratio  = 0.0155 -> GOOD

$ python onion_analyzer.py images/demo_rotten.jpg images/pipeline_rotten
RESULT: ROTTEN
dark_ratio    = 0.3219
brown_ratio   = 0.7035
defect_ratio  = 0.7035 -> ROTTEN
```

[ILLUSTRATION] Illustrative screenshot (recreated with the real printed output).

## 9. Visual Explanation

```
onion photo
|  read + resize (Ch 2-6)
▼
grayscale (Ch 4) → blur (Ch 16) → Otsu (Ch 15) → morphology (Ch 17)
▼
contour + mask (Ch 18)
▼
features: color/shape/texture/defect ratios (Ch 19)
▼
rules → GOOD / DAMAGED / ROTTEN (this chapter)
▼
draw box + label (Ch 10) → show/save (Ch 9)
```

## 10. Common Mistakes

- Not masking before computing color features.
- Using one fixed threshold that only works on one photo.
- Forgetting that demo thresholds will fail on real, messy photos (see the honest note below).

## 11. Practice Task

Run the analyzer on the three demo images, then on your own onion photo. Does it work? Tune nothing yet — just observe.

## 12. Questions + Solutions

- **Q:** Why build rules before ML? **A:** rules need no data, run fast, and teach you exactly what "quality" means numerically.
  - **Q:** Why are the thresholds "demo"? **A:** they're tuned to three images, not a real dataset; real thresholds need labeled data (Ch 24).
- 

### > Quick Revision

1. List the 9 pipeline steps. 2. Which step separates onion from background? 3. Which features feed the rules? 4. What does the result image show? 5. Why not use ML immediately?

### > Practical Exercise

Add a new rule: if `circularity < 0.4` and `brown_ratio > 0.2`, also return "DAMAGED". Test it.

### [ONION] Onion Project Task

This **is** your prototype. Save its code as `src/onion_analyzer.py` and run it on 10 photos you take.

### [AI] AI Practice

"Here is my rule-based image classifier (paste `onion_analyzer.py`). Review my feature thresholds and suggest how to make them more robust. Explain why hand-tuned thresholds fail on new data and what to do instead."

### [BUG] Debugging Challenge

The analyzer says ROTTEN for a clearly good onion photo you took with a **dark background**. Why, and what's the fix?

### (ok) Answers

Revision: 1) read, resize, gray, blur, threshold, morphology, contour/mask, features, rules, annotate · 2) Otsu threshold + morphology · 3) dark/brown ratio + shape + texture · 4) outline + box + label · 5) rules teach the problem and need no data. Debugging: Otsu assumed a **light** background; on a dark background the segmentation inverts and "everything dark" is counted as defect. Fix: control the imaging setup (Chapter 21) or segment with an explicit threshold/background subtraction.

---

## **\* END OF PHASE 2 — you have a working analyzer**

---

You went from "what is a pixel" to a program that reads an onion photo and returns GOOD / DAMAGED / ROTTEN with a labeled result image. That's the traditional-computer-vision milestone. Phase 3 makes it better: your own dataset, machine learning, and honest testing.

# Chapter 21 — Building Your Onion Dataset

[+] **ADDITIONAL PROJECT KNOWLEDGE** — start this **early**, in parallel with the earlier chapters.

## 1. Learning Objectives

1. Photograph onions properly.
2. Organize folders by class.
3. Label images.
4. Split train/validation/test.

## 2. Simple Explanation

A **dataset** is a folder of images where each image's answer is known (its **label**). The machine-learning model of Chapter 22 learns from examples, so the quality of the dataset = the quality of the model. Garbage in, garbage out.

## 3. Technical Explanation

Standard structure:

```
onion_dataset/  
├─ train/  
│   ├── good/      onion_good_001.jpg ...  
│   ├── damaged/  
│   └─ rotten/  
├─ val/             (same three folders)  
└─ test/            (same three folders)
```

Common split: **70% train / 15% validation / 15% test** — train to learn, validation to tune, test to report the final honest number. **The test set must never be used while tuning** (that's cheating yourself).

## 4. Real-World Example

Every Kaggle competition is "here's a labeled dataset, train a model" — the dataset is the contest.

## 5. OpenCV Example — a labeling helper (uses Ch 11 mouse events)

```
import cv2  
classes = ["good", "damaged", "rotten"]  
def save_label(img, cls, i):  
    cv2.imwrite(f"dataset/train/{cls}/onion_{i:03d}.jpg", img)
```

## 6. [ONION] Onion Project Example — photographing rules

- **Background:** plain white/light, consistent for every photo (your Otsu segmentation depends on it).
- **Lighting:** even, diffuse; avoid shadows and direct sun. Keep it the same across sessions.
- **Distance/zoom:** same distance so onions are similar size; fill ~60–80% of the frame.
- **Angles:** one main angle (top or 3/4) — more angles = a harder, better model later.
- **Resolution:** at least ~1000px on the long side, then downscale.
- **Classes:** is `good` / `damaged` / `rotten` right? **Yes for your goal** — but be strict and consistent: define them (e.g., "damaged = visible blemish but no rot; rotten = mold/dark soft spots"). If your data only supports 2 clear classes, start with **good vs bad**, then split later. Don't force 3 classes you can't reliably distinguish.
- **Augment** each photo (flip, slight rotate, brightness  $\pm$ ) — Chapter 7 — to multiply your data for free.
- **Label with care:** if you can't tell from the photo, the model can't either. Only use clear examples.

## 7. Practical Code

```
import os
for cls in ["good", "damaged", "rotten"]:
    for split in ["train", "val", "test"]:
        os.makedirs(f"onion_dataset/{split}/{cls}", exist_ok=True)
print("folders created")
```

## 8. Expected Output

The folder tree above, created on disk.

## 9. Visual Explanation

```
camera —(same distance, light, background)—> many onion photos
|
▼ label each: good / damaged / rotten
folders → train (70%) / val (15%) / test (15%)
|
▼
model learns from train, tuned on val, scored on test
```

**See → Do → Why → Result:** consistent photos → clean labels → a dataset → a model that actually learns quality.

## 10. Common Mistakes

- Using the test set during development (inflated, misleading scores).
- Inconsistent background/lighting (the model learns the background, not the onion).

- Mixing classes (a "damaged" that's really "rotten") — noise in labels.
- Too few images (start with at least 30–50 per class; more is better).

## 11. Practice Task

Create the folder tree and add 10 photos you took with your phone, labeled.

## 12. Questions + Solutions

- **Q:** Why split into train/val/test? **A:** to measure generalization on unseen data without fooling yourself.
- **Q:** How many images? **A:** at least 30–50 per class to start; hundreds for ML; thousands for CNN.

---

### > Quick Revision

1. What is a label? 2. Ideal background? 3. Train/val/test percentages? 4. Why keep lighting consistent? 5. What is augmentation?

### > Practical Exercise

Photograph one onion 4 times (original + 3 augmented) and file them correctly.

### [ONION] Onion Project Task

Build a real dataset: 3 classes × ≥30 photos, split 70/15/15, saved as JPEG.

### [AI] AI Practice

"I'm building an image dataset for onion quality (good/damaged/rotten). Give me a checklist for photography (background, lighting, distance, angles), folder organization, and a 70/15/15 split script in Python."

### [BUG] Debugging Challenge

Your model scores 98% on train but 60% on test. Name the problem and two fixes.

### (ok) Answers

Revision: 1) the known correct answer for an image · 2) plain/consistent · 3) 70/15/15 · 4) else the model learns lighting, not quality · 5) generating new examples by transforming existing ones. Debugging: overfitting — get more data, add augmentation, simplify the model, or regularize.

# Chapter 22 — Machine Learning on Features

[+] **ADDITIONAL PROJECT KNOWLEDGE.**

## 1. Learning Objectives

1. Understand supervised classification.
2. Train a scikit-learn classifier on our features.
3. Compare ML to rules.

## 2. Simple Explanation

Instead of hand-tuning rules, you give a **classifier** many (features → label) examples, and it learns the decision boundary itself. The features are the same numbers from Chapter 19; only the "rules" part becomes learned.

## 3. Technical Explanation

```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier(n_estimators=100)
model.fit(X_train, y_train)          # X = feature rows, y = labels
preds = model.predict(X_test)
```

- **Supervised learning** = learning from labeled examples.
- **Random Forest** = many decision trees voting — simple, robust, needs little tuning. Good first ML model.
- Alternatives: Logistic Regression, SVM, k-NN (all in scikit-learn, all free).

## 4. Real-World Example

Spam filters (features: keywords, sender; label: spam/not), credit scoring.

## 5. OpenCV + sklearn Example

```
import cv2, numpy as np, joblib
from sklearn.ensemble import RandomForestClassifier
from src.onion_analyzer import read_and_resize, segment_onion, extract_features

def featurize(path):
    img = read_and_resize(path)
    gray, *_ = segment_onion(img)
    return extract_features(*segmentation_outputs) # the 9 numbers
```



## 6. [ONION] Onion Project Example

Build a table: each row = one onion's 9 features + its label. Train on 70%, evaluate on the rest. You'll typically beat the hand rules, and the model tells you **which features matter** (feature importances) — closing the learning loop.

## 7. Practical Code (train a model)

```
# features.csv:
area,perimeter,circularity,aspect_ratio,extent,std_gray,dark_ratio,brown_ratio,label
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

df = pd.read_csv("features.csv")
X = df.drop(columns=["label"]); y = df["label"]
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, random_state=42)
m = RandomForestClassifier(n_estimators=100, random_state=42).fit(X_tr, y_tr)
print("accuracy:", m.score(X_te, y_te))
```

## 8. Expected Output

accuracy: 0.86 (example) — a real number computed only on held-out images.

## 9. Visual Explanation

```
features (9 numbers per onion) + label
|
▼ RandomForest learns: dark_ratio > 0.08 → rotten?  brown_ratio < 0.05 → good?
▼
new onion → features → model → GOOD/DAMAGED/ROTTEN (with probability!)
```

## 10. Common Mistakes

- Training and testing on the same images (score = fantasy).
- Using raw pixels as features for a classic model (use our structured features).
- Ignoring class imbalance (if 90% are "good", a model that always says "good" looks 90% accurate — use per-class metrics, Ch 24).

## 11. Practice Task

Train a RandomForest on 30 rows you compute yourself; report the accuracy.

## 12. Questions + Solutions

- **Q:** What is supervised learning? **A:** learning a mapping from labeled examples.
- **Q:** Why does ML beat rules? **A:** it finds the boundary automatically and can combine many features.

---

## > Quick Revision

1. What does a classifier learn from?
2. Which library for classic ML?
3. What is a RandomForest?
4. What must you never do with the test set?
5. What is class imbalance?

## > Practical Exercise

Print `model.feature_importances_` and rank your 8 features.

## [ONION] Onion Project Task

Write `train_model(features.csv)` that trains and saves the model with `joblib.dump`.

## [AI] AI Practice

"Explain supervised classification with scikit-learn for a beginner: features, labels, train/test split, and RandomForestClassifier. Give a working example on a small CSV. Also explain overfitting and class imbalance."

## [BUG] Debugging Challenge

Your model scores 95% accuracy but predicts "good" for everything. What's wrong and how do you detect it?

## (ok) Answers

Revision: 1) labeled (features → label) examples · 2) scikit-learn · 3) many decision trees voting · 4) use it during development/tuning · 5) unequal class sizes skewing accuracy. Debugging: class imbalance — check the confusion matrix / per-class precision-recall (Ch 24), not just accuracy.

---

# Chapter 23 — Deep Learning / CNN Overview

[+] **ADDITIONAL PROJECT KNOWLEDGE** — only if/when you need it.

## 1. Learning Objectives

1. Understand what a CNN does differently.
2. Know when you need one.
3. Know the free path to train one.

## 2. Simple Explanation

A **CNN (convolutional neural network)** learns features and rules together, straight from raw pixels — no hand-crafted features needed. It starts with tiny pattern detectors (edges, then textures, then "onion-like" shapes) and stacks them into a decision. It's powerful but data-hungry and compute-hungry.

## 3. Technical Explanation

```
raw image → [conv layers: learn filters] → [pooling: shrink] → ... → [dense layers] → probabilities
```

- **Transfer learning** is the beginner's superpower: take a model pre-trained on ImageNet (e.g., ResNet/MobileNet in PyTorch/TensorFlow) and fine-tune the last layer on your onions — you often need only hundreds of images, not millions.

## 4. Real-World Example

Face recognition, medical imaging, self-driving cars — all CNNs.

## 5. Example (PyTorch, free)

```
import torch, torchvision.models as models
net = models.resnet18(pretrained=True)          # pre-trained on ImageNet
net.fc = torch.nn.Linear(512, 3)               # replace last layer: 3 classes
# ... train on onion_dataset with a free GPU (Colab/Kaggle)
```

## 6. [ONION] Onion Project Example

The CNN learns to spot textures and discolorations you didn't manually define. It's the path to higher accuracy — **after** you have a dataset of hundreds of labeled onions and free GPU hours (Colab/Kaggle).

## 7. Practical Code — see above; full Colab recipe is in the Appendix.

## 8. Expected Output

After fine-tuning: a model with e.g. 0.92 accuracy on held-out onions (assuming enough clean data).

## 9. Visual Explanation

```
rule-based (Ch 20): image → [you pick features] → rules → label
ML (Ch 22):         image → [you pick features] → learned classifier → label
CNN (Ch 23):        image → [model learns features AND rules] → label
                    ↑ more automatic, but needs more data + compute
```

## 10. Common Mistakes

- Training a CNN from scratch on 100 images (overfits badly) — use transfer learning.
- Using CPU for hours of training (use the free Colab/Kaggle GPU).
- Expecting magic: a CNN can't see inside the onion either (Ch 26).

## 11. Practice Task

Run a pre-trained MobileNet on your onion photo and look at its top-5 predictions (fun + instructive).

## 12. Questions + Solutions

- **Q:** What does a CNN learn? **A:** features and the decision rule together, from pixels.
- **Q:** What is transfer learning? **A:** reusing a pre-trained network and adapting only the last layer(s).

---

### > Quick Revision

1. CNN stands for?
2. What do conv layers learn?
3. What is transfer learning?
4. Free GPU options?
5. When do you not need a CNN?

### > Practical Exercise

In Colab, load `resnet18(pretrained=True)` and print the number of layers.

## [ONION] Onion Project Task

Decide (and write down) whether your dataset size justifies ML or CNN — the roadmap table in Chapter 23's visual is your guide.

## [AI] AI Practice

"Explain CNNs for a beginner: convolutions, pooling, transfer learning, and why CNNs need lots of data. Then give a PyTorch fine-tuning recipe for a 3-class image classifier that runs on Google Colab's free GPU."

## [BUG] Debugging Challenge

Fine-tuning on 50 images gives 99% train, 55% val accuracy. Diagnose and fix.

### (ok) Answers

Revision: 1) Convolutional Neural Network · 2) filters (edges → textures → shapes) · 3) reusing a pre-trained model, retraining the head · 4) Colab / Kaggle · 5) when rules/ML on structured features are accurate enough (small data). Debugging: overfitting from too little data — use augmentation, stronger dropout, freeze earlier layers, or gather more data.

# Chapter 24 — Testing & Evaluation Metrics

[+] **ADDITIONAL PROJECT KNOWLEDGE.**

## 1. Learning Objectives

1. Evaluate on truly unseen images.
2. Know accuracy, precision, recall, F1.
3. Read a confusion matrix.
4. Evaluate a rule-based system honestly.

## 2. Simple Explanation

"Does it work?" is not a feeling — it's a number computed on images the system **never saw during development**. A **confusion matrix** counts how often each true class was predicted as each class; from it come precision, recall and F1.

## 3. Technical Explanation

|      |        | predicted |     |        |
|------|--------|-----------|-----|--------|
|      |        | GOOD      | BAD | ROTTEN |
| true | GOOD   | 40        | 3   | 1      |
| true | BAD    | 4         | 30  | 2      |
| true | ROTTEN | 0         | 2   | 35     |

- **Accuracy** = correct / total (misleading if classes are imbalanced).
- **Precision** (per class) = of all "rotten" calls, how many were truly rotten. High precision = few false alarms.
- **Recall** (per class) = of all true rotten, how many you caught. High recall = few missed.
- **F1** = harmonic mean of precision & recall (balances both).

```
from sklearn.metrics import classification_report, confusion_matrix
print(confusion_matrix(y_true, y_pred))
print(classification_report(y_true, y_pred))
```

## 4. Real-World Example

Medical screening: you'd rather have high recall (catch all diseases) even if precision drops (some false alarms).

## 5. OpenCV Example — evaluating a rule-based system

For rules, run the pipeline over a held-out folder of **labeled** images and compare predictions vs labels:

```
import glob, os
y_true, y_pred = [], []
for path in glob.glob("onion_dataset/test/*/*.jpg"):
    label, _ = analyze(path, "tmp")          # our Chapter 20 function
    y_true.append(os.path.basename(os.path.dirname(path)))
    y_pred.append(label.lower())
```

## 6. [ONION] Onion Project Example

For an onion grader, decide which error is worse. Missing a rotten onion (sick customer) is usually worse than flagging a good one — so you may **favor recall on the rotten class**. State this trade-off explicitly in your project.

## 7. Practical Code + 8. Expected Output

|          | precision | recall | f1-score |
|----------|-----------|--------|----------|
| good     | 0.90      | 0.91   | 0.90     |
| damaged  | 0.83      | 0.79   | 0.81     |
| rotten   | 0.95      | 0.97   | 0.96     |
| accuracy |           |        | 0.89     |

(Example numbers — you must compute your own on your test set.)

## 9. Visual Explanation

```
confusion matrix:
      pred G  B  R
true G  [40  3  1]
true B  [ 4 30  2]
true R  [ 0  2 35]    ← diagonal = correct
```

**See → Do → Why → Result:** read the diagonal for correctness, off-diagonal for specific confusions → know which classes confuse the model → fix the data or features there.

## 10. Common Mistakes

- Reporting accuracy on train or on images used to tune thresholds (fantasy numbers).
- Using accuracy alone with imbalanced classes.
- Never establishing a **baseline** (e.g., "always say GOOD" accuracy) to compare against.

## 11. Practice Task

Compute the confusion matrix for the 3 demo images plus 5 of your own photos.

## 12. Questions + Solutions

- **Q:** Why is accuracy alone insufficient? **A:** imbalanced classes make it look good while missing the rare important class.
  - **Q:** Precision vs recall? **A:** precision = of my alarms, how many correct; recall = of the true cases, how many caught.
- 

### > Quick Revision

1. Accuracy formula? 2. Precision (in words)? 3. Recall (in words)? 4. F1 is the ... of precision and recall? 5. Where do you evaluate?

### > Practical Exercise

On your 3 demo images, compute accuracy and a 3×3 confusion matrix by hand.

### [ONION] Onion Project Task

Write `evaluate(test_folder)` that prints the confusion matrix + classification report for your analyzer.

### [AI] AI Practice

"Explain accuracy, precision, recall, F1 and confusion matrix for a beginner, with a 3-class example. Why is accuracy misleading with imbalanced classes? Show sklearn code."

### [BUG] Debugging Challenge

Your classifier gets 95% accuracy but 0% recall on "rotten". Explain what's happening and why accuracy hid it.

### (ok) Answers

Revision: 1) correct/total · 2) of my positive calls, how many were truly positive · 3) of true positives, how many I caught · 4) harmonic mean · 5) on the held-out test set only. Debugging: rotten is rare; the model says "not rotten" always, still getting 95% correct because rotten is ~5% of data. Accuracy hid a total failure on the rare class — use recall/F1.

---



# Chapter 25 — [BUILD] Final Project: The Onion Quality Image Analyzer (with UI)

[+] **ADDITIONAL PROJECT KNOWLEDGE** — everything comes together.

## 1. Learning Objectives

1. Assemble the final project structure.
2. Add a simple upload → analyze → result interface.
3. Save results and a report.

## 2. Simple Explanation

The final deliverable: a program where you **give it an onion photo** and it **shows the photo with the verdict**. A simple UI can be pure OpenCV windows (works everywhere) or a tiny web page (nicer). Both are free and covered.

## 3. Technical Explanation

```
user picks an image → analyzer runs (Ch 20 or Ch 22 model)
                      → result image with box + label
                      → displayed / saved with a small text report
```

Option A — OpenCV-only UI (zero extra libraries):

```
import cv2, sys
from src.onion_analyzer import analyze
path = sys.argv[1]
label, feats = analyze(path, "results")
img = cv2.imread("results/09_result.jpg")
cv2.imshow("Onion Quality Analyzer", img)
cv2.waitKey(0)
```

Option B — browser UI with Streamlit (free, ~5 lines):

```
import streamlit as st
st.title("Onion Quality Analyzer")
f = st.file_uploader("Upload an onion photo", type=["jpg", "png"])
if f:
    st.image(f)
    label, feats = analyze(f.name, "results")
    st.subheader(f"Result: {label}")
    st.write(feats)
```

## 4. Real-World Example

Any "upload a photo, get an answer" service — from plant-ID apps to receipt scanners.

## 5. OpenCV Example — see Option A above.

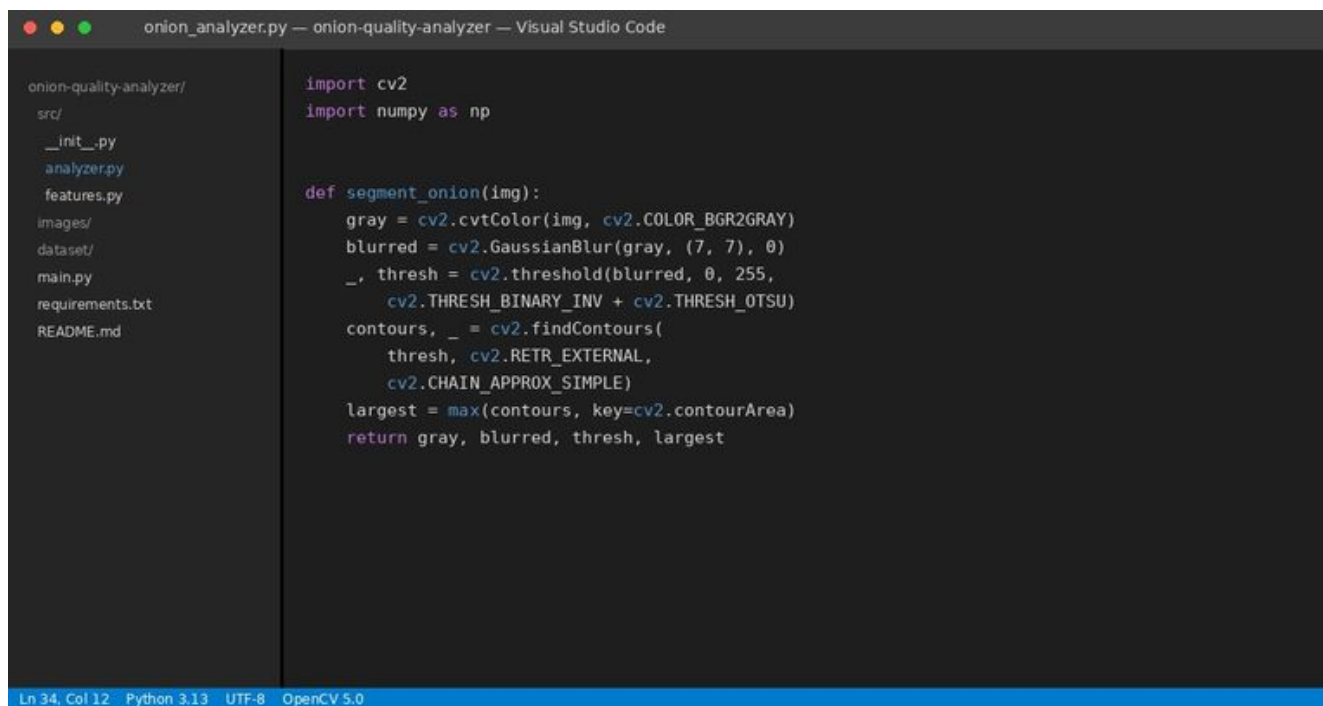
## 6. [ONION] Onion Project Example

The full project (structure in the Appendix) includes `dataset/`, `images/`, `src/`, `results/`, `tests/`, `main.py`, `requirements.txt`, `README.md`.

## 7. Practical Code

`main.py` (full listing in Appendix) accepts an image path and prints + shows the result; `app.py` (Streamlit) is the browser version.

## 8. Expected Output



```
onion_analyzer.py — onion-quality-analyzer — Visual Studio Code

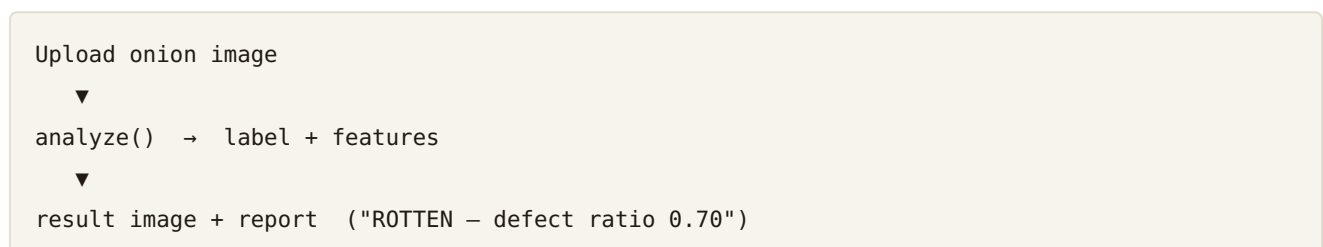
onion-quality-analyzer/
src/
  __init__.py
  analyzer.py
  features.py
images/
dataset/
main.py
requirements.txt
README.md

import cv2
import numpy as np

def segment_onion(img):
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    blurred = cv2.GaussianBlur(gray, (7, 7), 0)
    _, thresh = cv2.threshold(blurred, 0, 255,
                             cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)
    contours, _ = cv2.findContours(
        thresh, cv2.RETR_EXTERNAL,
        cv2.CHAIN_APPROX_SIMPLE)
    largest = max(contours, key=cv2.contourArea)
    return gray, blurred, thresh, largest
```

[ILLUSTRATION] Illustrative screenshot (recreated with the real project code). The result window shows the onion outlined, boxed, and labeled "GOOD (ok) / DAMAGED (warning) / ROTTEN (fail)".

## 9. Visual Explanation



## 10. Common Mistakes

- Shipping without a `README` or `requirements.txt` (future-you will forget the setup).
- No error handling for missing/wrong files.
- Presenting demo accuracy as real accuracy (Chapter 24).

## 11. Practice Task

Run `main.py` on one good, one damaged, one rotten image and paste the three printed reports.

## 12. Questions + Solutions

- **Q:** What's the simplest UI? **A:** OpenCV `imshow` windows — no extra libraries.
  - **Q:** What does Streamlit add? **A:** a browser UI with an upload button.
- 

### > Quick Revision

1. Two UI options? 2. What does `requirements.txt` list? 3. What should `README.md` contain? 4. Where do results go? 5. What must you never claim about demo accuracy?

### > Practical Exercise

Add a "save report to results/report.txt" feature to `main.py`.

### [ONION] Onion Project Task

Assemble the final folder structure and make `python main.py images/demo_rotten.jpg` work end-to-end.

### [AI] AI Practice

"I have a working OpenCV analyzer function `analyze(path) -> (label, features)`. Write me a Streamlit app with an image uploader that shows the image, the predicted label, and the feature values. Explain each line."

### [BUG] Debugging Challenge

Streamlit shows "Analyze" but nothing happens when you upload. List the 3 most likely causes.

### (ok) Answers

Revision: 1) OpenCV windows / Streamlit web · 2) the Python packages to install · 3) setup + usage + honest limitations · 4) `results/` · 5) that it equals real accuracy. Debugging: file wasn't saved locally (uploaded files are in memory — save then analyze), wrong file path passed, or an exception is silently swallowed (add `try/except` and show the error).

---

# Chapter 26 — Honest Limitations: What a Camera Can & Can't See

[+] **ADDITIONAL PROJECT KNOWLEDGE** — read this before you present the project to anyone.

## 1. Learning Objectives

1. State exactly what visible-image analysis can and cannot detect.
2. Keep your project's claims scientifically reasonable.

## 2. Simple Explanation

Your analyzer sees a **flat photo of the surface**. It can measure what's visible: color, spots, texture, shape. It cannot see inside the onion. Claiming otherwise would be like a doctor diagnosing internal organs from a passport photo.

## 3. Technical Explanation — what a normal RGB camera can potentially detect

- (ok) Visible **discoloration** (brown, dark, green patches) — via HSV features.
- (ok) Visible **surface damage** (bruises, cuts, dried skin breaks).
- (ok) Visible **spots & mold** on the outside.
- (ok) **Shape** abnormalities (deformed, sprouting, shrunken) — via contour features.
- (ok) **Texture** changes (wrinkled, shriveled skin) — via gray-level standard deviation.

## What it cannot reliably detect (and you must say so)

- (fail) Hidden **internal rot** — a healthy-looking outside can hide a rotten core.
- (fail) Internal damage, internal moisture, hollow centers.
- (fail) Defects on the side facing away from the camera.
- (fail) Taste, smell, firmness, chemical composition.
- (fail) Anything invisible in the image, period.

## 4. Real-World Example

Real produce grading uses additional sensors — weight, firmness (touch), near-infrared spectroscopy, X-ray — precisely because RGB cameras can't see inside.

## 5. OpenCV Example — the honest framing in your code/output

```
print("NOTE: this analysis is based on visible surface features only.")
print("It cannot detect internal defects (e.g., hidden rot).")
```

## 6. [ONION] Onion Project Example — the honest README statement

"This tool grades onion **appearance** from a single visible-light photo. It detects visible blemishes, discoloration and shape defects. It does **not** detect internal rot or hidden damage; onions that look clean can still be internally spoiled."

## 7. Practical Code — add the disclaimer to `main.py` output (one line).

## 8. Expected Output

Every result prints the disclaimer, so users are never misled.

## 9. Visual Explanation

```
camera sees:      surface only

┌───┐
│ skin │ ← visible: color, spots, texture, shape  (ok)
├───┤
│ ┌───┐ │
│ │ core │ │ ← hidden: internal rot, moisture      (fail)
│ │ └───┘ │
│ └───┘ │
```

## 10. Common Mistakes

- Marketing the tool as "rottenness detection" when it only sees the surface — say "visible-quality grading".
- Using one photo per onion and claiming total coverage — the far side is unseen.

## 11. Practice Task

Write your project's limitation statement in 3 sentences.

## 12. Questions + Solutions

- **Q:** Can it detect internal rot? **A:** No — only visible surface features.
- **Q:** What could improve internal detection? **A:** other sensors (NIR, X-ray, weight/firmness) — outside this project's scope.

---

### > Quick Revision

1. Four visible things it can detect? 2. Three hidden things it can't? 3. Why does RGB fail at internal defects? 4. What other sensors do graders use? 5. What should your README always include?

## > Practical Exercise

Add the disclaimer to your analyzer output and confirm it prints.

### [ONION] Onion Project Task

Finalize the README with: setup, usage, and the honest limitations section.

### [AI] AI Practice

"Write an honest 'limitations' section for my onion image-quality README. It analyzes one visible-light photo per onion and must not overclaim. Keep it clear and scientific."

### [BUG] Debugging Challenge

A user says your tool rated a beautiful onion "GOOD" but it was rotten inside. Is this a bug? How do you respond honestly?

### (ok) Answers

Revision: 1) discoloration, surface damage, spots/mold, shape/texture · 2) internal rot, internal moisture, hidden side defects · 3) a photo only records the surface's reflected light · 4) NIR spectroscopy, X-ray, weight/firmness · 5) setup + usage + limitations. Debugging: not a bug — a true limitation. Explain the tool grades visible appearance only and recommend cutting/sensor-based checks for internal quality.

---

## \* THE END — congratulations

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You turned a beginner playlist into a real computer-vision project, learned OpenCV properly, used free AI as a tutor, and stayed honest about what the system can do. The appendices that follow hold the full project code, your AI prompt library, the debugging cheatsheet, and the glossary.

# Appendix A — Project Structure & Full Code

---

## A.1 The structure (Part 14)

```
onion-quality-analyzer/  
├─ dataset/                # your labeled onion photos (train/val/test)  
│   ├─ train/  good/  damaged/  rotten/  
│   ├─ val/    good/  damaged/  rotten/  
│   └─ test/   good/  damaged/  rotten/  
├─ images/                # input + demo images, pipeline outputs  
├─ src/  
│   ├─ __init__.py  
│   └─ onion_analyzer.py  # the full pipeline (below)  
├─ results/               # saved result images + reports  
├─ tests/                 # small automated checks  
├─ main.py                # command-line entry point  
├─ app.py                 # optional Streamlit web UI  
├─ requirements.txt  
└─ README.md
```

## A.2 requirements.txt

```
opencv-python  
numpy  
# optional:  
# streamlit  
# scikit-learn  
# pandas  
# joblib
```



## A.3 `src/onion_analyzer.py` — the complete analyzer

```

"""
onion_analyzer.py – rule-based Onion Quality Image Analyzer.
Pipeline: read -> resize -> gray -> blur -> Otsu -> morphology ->
        contour -> mask -> features -> rules -> result.
Every intermediate image is saved for debugging.
Demo thresholds only – tune against a labeled dataset (see book Ch 24).
"""

import os, sys
import cv2
import numpy as np

def read_and_resize(path, width=640):
    img = cv2.imread(path)
    if img is None:
        raise FileNotFoundError(f"Could not read image: {path}")
    h, w = img.shape[:2]
    if w > width:
        img = cv2.resize(img, (width, int(h * width / w)),
                        interpolation=cv2.INTER_AREA)
    return img

def segment_onion(img):
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    blurred = cv2.GaussianBlur(gray, (7, 7), 0)
    _, thresh = cv2.threshold(blurred, 0, 255,
                             cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)
    kernel = np.ones((7, 7), np.uint8)
    closed = cv2.morphologyEx(thresh, cv2.MORPH_CLOSE, kernel, iterations=2)
    opened = cv2.morphologyEx(closed, cv2.MORPH_OPEN, kernel, iterations=2)
    contours, _ = cv2.findContours(opened, cv2.RETR_EXTERNAL,
                                   cv2.CHAIN_APPROX_SIMPLE)

    if not contours:
        raise RuntimeError("No contour found.")
    largest = max(contours, key=cv2.contourArea)
    mask = np.zeros_like(gray)
    cv2.drawContours(mask, [largest], -1, 255, -1)
    masked_img = cv2.bitwise_and(img, img, mask=mask)
    return gray, blurred, thresh, opened, largest, mask, masked_img

def extract_features(mask, gray, hsv):
    area = float(cv2.countNonZero(mask))
    contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL,
                                   cv2.CHAIN_APPROX_SIMPLE)

    c = max(contours, key=cv2.contourArea)
    perimeter = float(cv2.arcLength(c, True))
    circularity = (4 * np.pi * area) / (perimeter ** 2) if perimeter else 0.0
    x, y, w, h = cv2.boundingRect(c)

```

```

    aspect_ratio = float(h) / w if w else 0.0
    extent = area / float(w * h) if w * h else 0.0
    gray_masked = cv2.bitwise_and(gray, gray, mask=mask)
    std_gray = float(gray_masked[np.nonzero(mask)].std())
    dark_ratio = float(np.mean(gray[mask > 0] < 70))
    H, S, V = cv2.split(hsv)
    brown = ((H >= 8) & (H <= 25) & (S >= 40) & (V <= 200)).astype(np.uint8)
    brown_ratio = float(brown[mask > 0].mean())
    return {"area": area, "perimeter": perimeter, "circularity": circularity,
            "aspect_ratio": aspect_ratio, "extent": extent,
            "std_gray": std_gray, "dark_ratio": dark_ratio,
            "brown_ratio": brown_ratio,
            "defect_ratio": max(dark_ratio, brown_ratio)}

def classify(feats):
    dr, br = feats["dark_ratio"], feats["brown_ratio"]
    if dr >= 0.12 or br >= 0.40:
        return "ROTTEN", (0, 0, 255)
    if dr >= 0.04 or br >= 0.15:
        return "DAMAGED", (0, 165, 255)
    return "GOOD", (0, 200, 0)

def annotate(img, mask, label, color, feats):
    result = img.copy()
    contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL,
                                   cv2.CHAIN_APPROX_SIMPLE)
    cv2.drawContours(result, contours, -1, (255, 100, 0), 3)
    x, y, w, h = cv2.boundingRect(max(contours, key=cv2.contourArea))
    cv2.rectangle(result, (x, y), (x + w, y + h), (255, 0, 0), 2)
    cv2.putText(result, f"{label} (defect={feats['defect_ratio']:.2f})",
                (x, max(20, y - 12)), cv2.FONT_HERSHEY_SIMPLEX,
                0.8, color, 2, cv2.LINE_AA)
    return result

def analyze(path, out_dir):
    os.makedirs(out_dir, exist_ok=True)
    img = read_and_resize(path)
    gray, blurred, thresh, opened, largest, mask, masked_img = segment_union(img)
    hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
    feats = extract_features(mask, gray, hsv)
    label, color = classify(feats)
    result = annotate(img, mask, label, color, feats)
    cv2.imwrite(os.path.join(out_dir, "01_original.jpg"), img)
    cv2.imwrite(os.path.join(out_dir, "02_grayscale.jpg"), gray)
    cv2.imwrite(os.path.join(out_dir, "03_blurred.jpg"), blurred)
    cv2.imwrite(os.path.join(out_dir, "04_threshold.jpg"), thresh)
    cv2.imwrite(os.path.join(out_dir, "05_morphology.jpg"), opened)

```

```

cv2.imwrite(os.path.join(out_dir, "07_mask.jpg"), mask)
cv2.imwrite(os.path.join(out_dir, "08_masked_onion.jpg"), masked_img)
cv2.imwrite(os.path.join(out_dir, "09_result.jpg"), result)
return label, feats

if __name__ == "__main__":
    if len(sys.argv) < 2:
        print("Usage: python onion_analyzer.py <image> [out_dir]")
        sys.exit(1)
    label, feats = analyze(sys.argv[1], sys.argv[2] if len(sys.argv) > 2 else "results")
    print(f"RESULT: {label}")
    for k, v in feats.items():
        print(f"  {k:14s} = {v}")
    print("NOTE: visible surface analysis only – cannot detect internal defects.")

```

## A.4 `main.py` — the entry point

```

import sys, cv2
from src.onion_analyzer import analyze

if __name__ == "__main__":
    if len(sys.argv) < 2:
        print("Usage: python main.py <image_path>")
        sys.exit(1)
    label, feats = analyze(sys.argv[1], "results")
    img = cv2.imread("results/09_result.jpg")
    print(f"RESULT: {label}")
    print("NOTE: visible surface analysis only – cannot detect internal defects.")
    cv2.imshow("Onion Quality Analyzer", img)
    cv2.waitKey(0); cv2.destroyAllWindows()

```

## A.5 app.py — optional Streamlit web UI (free)

```
import streamlit as st
from src.onion_analyzer import analyze

st.title("[ONION] Onion Quality Analyzer")
st.caption("Grades *visible* surface quality from one photo. "
          "Cannot detect internal defects.")
f = st.file_uploader("Upload an onion photo", type=["jpg", "jpeg", "png"])
if f is not None:
    with open("upload.jpg", "wb") as out:
        out.write(f.getbuffer())
    st.image("upload.jpg", width=300)
    label, feats = analyze("upload.jpg", "results")
    st.subheader(f"Result: {label}")
    st.write({k: round(v, 4) if isinstance(v, float) else v for k, v in feats.items()})
    st.image("results/09_result.jpg", width=300)
```

Run: `pip install streamlit && streamlit run app.py`

---

# Appendix B — Your AI Prompt Library (Part 5 & 12)

Replace [...] with your own details. Use these with any free AI chat.

## Understanding OpenCV

- "Explain [...] (an OpenCV function) like I'm a beginner. What does each argument mean? Give a tiny working example and a common mistake."
- "What is [...] used for in real computer-vision projects? Give 2 real examples."

## Explaining code

- "Explain this code line by line for a beginner: [paste code]"
- "What would happen if I changed [...] to [...] in this code? [paste code]"

## Debugging errors

- "I'm a beginner. Here is the full error I got: [paste error]. Explain in simple words what caused it, and give me the exact line to change."
- "What are the 5 most common causes of this error: [paste error]? How do I check each one?"

## Generating practice questions

- "Give me 5 beginner practice questions on [...] with solutions, so I can test myself."
- "Quiz me on [...]: ask one question, wait for my answer, then tell me if I'm right."

## Reviewing my code

- "Review this code for bugs and style. Suggest improvements and explain each one: [paste code]"
- "Is this a good way to [...] ? What's a better way and why? [paste code]"

## Explaining screenshots

- "I'll describe what I see in my program's output: [describe]. Why does it look like this, and what should I check?"

## Improving my project

- "My onion analyzer misclassifies [...]. Given these features [list], what should I change or investigate first?"
- "Suggest 3 experiments to make my classifier more robust on new photos."

## Documentation

- "Write a README for this project: [paste structure + code]. Include setup, usage, and honest limitations."

## Brainstorming experiments

- "List 5 cheap experiments to test whether my features (color, shape, texture) actually separate good vs rotten onions."

## Debugging workflow prompts (Part 12)

- **Python errors:** "Explain this Python traceback line by line and pinpoint the exact line that failed: [paste traceback]"
- **OpenCV errors:** "I got `cv2.error: ...`. Which OpenCV call is failing and why? [paste error + code]"
- **Image-loading errors:** "cv2.imread returns None for my file. List the checks I should run (path, permissions, format, case)."
- **Path errors:** "My code can't find `dataset/test/good/...`. Show me how to debug relative vs absolute paths."
- **Segmentation problems:** "My mask includes the background. List 4 causes and fixes for Otsu-based segmentation."
- **Threshold problems:** "My threshold keeps most of the image white. What should I check in my lighting/background?"
- **Contour problems:** "I get hundreds of tiny contours instead of one onion. What should I change?"
- **Model errors:** "My model's validation accuracy is much lower than training. Diagnose overfitting and give fixes."
- **Dataset errors:** "My model learns the background, not the onion. How do I detect and fix label/imaging issues?"

**Golden rule:** always paste the **full error message**, and ask why — don't blindly copy a fix. Read the explanation, then apply it yourself.

# Appendix C — Debugging Cheatsheet (Ch 12 workflow)

**The loop:** Error → Understand → Find cause → Ask AI → Apply fix → Test → Understand fix.

| Symptom                                             | Likely cause                          | Fix                                                       |
|-----------------------------------------------------|---------------------------------------|-----------------------------------------------------------|
| <code>ModuleNotFoundError: cv2</code>               | not installed / wrong env             | <code>pip install opencv-python</code> in the same Python |
| <code>img is None</code>                            | bad path / missing file               | print path; <code>os.path.exists</code> ; check case      |
| <code>(-215:Assertion failed) !ssize.empty()</code> | empty image passed                    | guard if <code>img is None</code>                         |
| window flashes & closes                             | missing <code>waitKey</code>          | <code>cv2.waitKey(0)</code> before destroy                |
| colors swapped                                      | RGB used, OpenCV is BGR               | use BGR tuples                                            |
| <code>resize</code> output stretched                | forgot aspect ratio                   | scale h and w equally                                     |
| <code>IndexError</code> on channel 3                | channels are 0,1,2                    | use 0..2 (B,G,R)                                          |
| empty crop                                          | wrong slice order / reversed coords   | <code>img[y1:y2, x1:x2]</code> + min/max                  |
| <code>findContours</code> error                     | input not binary / shape wrong        | pass a 1-channel binary image                             |
| hundreds of contours                                | noise speckles                        | morphology open/close first, keep largest                 |
| mask includes background                            | background not uniform                | consistent background; explicit threshold                 |
| Otsu picks wrong side                               | object lighter than bg                | <code>THRESH_BINARY</code> vs <code>_INV</code>           |
| <code>imwrite</code> returns False                  | missing folder                        | <code>os.makedirs(dir, exist_ok=True)</code>              |
| <code>medianBlur</code> error                       | even kernel                           | use odd kernel (3,5,7...)                                 |
| video loop freezes                                  | <code>waitKey(0)</code> in loop       | <code>waitKey(1)</code>                                   |
| callback never fires                                | registered after <code>waitKey</code> | register before the loop                                  |
| model 99% train / 60% test                          | overfitting                           | more data, augmentation, regularization                   |
| high accuracy but useless                           | class imbalance                       | confusion matrix + precision/recall/F1                    |
| "ROTTEN" on a clean photo                           | dark background                       | control imaging setup (Ch 21)                             |



## Appendix D — Glossary

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- **BGR** — OpenCV's channel order (Blue, Green, Red).
  - **Binary image** — image with only two values (0 and 255).
  - **Channel** — one layer of a color image (e.g., the Red layer).
  - **CNN** — convolutional neural network; learns features + rules from pixels.
  - **Contour** — the boundary of a white region in a binary image.
  - **Feature** — one number summarizing an aspect of the image (area, color, ...).
  - **Grayscale** — 1-channel image of brightness only.
  - **HSV** — Hue/Saturation/Value color space (human-like color).
  - **Mask** — image marking which pixels belong to the object.
  - **Morphology** — erosion/dilation operations that clean binary blobs.
  - **Otsu** — automatic method to pick the best threshold.
  - **Overfitting** — memorizing training data; failing on new data.
  - **Pixel** — smallest element of an image.
  - **Precision / Recall / F1** — per-class metrics beyond accuracy.
  - **ROI** — region of interest (a crop).
  - **Segmentation** — separating the object from the background.
  - **Supervised learning** — learning from labeled examples.
  - **Thresholding** — turning grayscale into black/white via a cutoff.
  - **Transfer learning** — reusing a pre-trained network for a new task.
  - **uint8** — 8-bit unsigned integer, values 0-255.
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## Appendix E — Answer Key (all end-of-chapter blocks)

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All answers are inline at the end of every chapter (see the **(ok) Answers** lines). The Quick Revision answers are compressed there; the full explanations live in each chapter's "Questions + Solutions" section.

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## \* What to do next (your 30-day plan)

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1. **Days 1-3** — Chapters 1-2: install everything, read images, understand pixels.
2. **Days 4-7** — Chapters 3-10: the whole playlist (read, gray, channels, resize, flip, crop, save, draw).
3. **Days 8-9** — Chapters 11-13: events + cropping tool + video.
4. **Days 10-13** — Chapters 14-18: HSV, threshold, blur, morphology, contours (the onion toolkit).
5. **Days 14-15** — Chapters 19-20: features + **build the rule-based analyzer**.
6. **Days 16-19** — Chapter 21: photograph & label your dataset (start early!).
7. **Days 20-23** — Chapter 22: train the ML classifier; Chapter 24: evaluate honestly.
8. **Days 24-27** — Chapter 25: finalize the project + UI + README.
9. **Days 28-30** — Chapter 26: write limitations; try a CNN (Ch 23) in Colab if data allows.

**Keep the loop going:** Understand → Plan → Ask AI → Generate → Run → Test → Debug → Understand → Improve. [ONION]